

Gender-biased Technology, Intra-household Bargaining, and Fertility

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Abstract

This paper examines how gender-biased technological change affects fertility in a developing economy through an underexplored channel: intra-household bargaining. We develop a collective household model with heterogeneous spousal preferences that decomposes the fertility response to a male-biased labor-demand shock into an opportunity-cost, an income, and a bargaining channel, and show that only the bargaining channel varies with the spousal preference gap and with gender norms. We take the model to China—the world’s largest installer of industrial robots, whose total fertility rate has continued to fall despite repeated relaxations of birth quotas. Using a Bartik shift-share instrument that exploits cross-prefecture variation in robot exposure, we find that a one-standard-deviation increase reduces fertility by about 3 percent. The evidence points strongly to the bargaining channel: robot exposure raises wives’ relative earnings and measured bargaining power, and the fertility response is larger where spousal preference gaps are wider and traditional gender norms stronger, shifting realized fertility toward women’s systematically lower preferences. The opportunity-cost channel cannot explain the decline, and although household income losses contribute to it, they cannot account for it on their own.

Keywords: Robots; gender-biased technology; fertility; intra-household bargaining; social norms

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1 Introduction

Technological change is rarely gender-neutral. From household appliances that freed women’s time (Greenwood et al., 2005) to the contraceptive pill that expanded their careers (Goldin and Katz, 2002; Bailey, 2006), transformative technologies reshape labor markets along gender lines, and, through that, fertility. The standard accounts of this link mainly run through two channels: an opportunity-cost channel, in which changes in women’s wages alter the shadow price of childcare time (Becker, 1960; Mincer, 1963; Schultz, 1985; Galor and Weil, 1996; Jakobsen et al., 2025), and an income channel, in which changes in household resources alter family size and child quality (Becker and Lewis, 1973; Jones et al., 2010; Galor and Weil, 2000).

However, a third channel—bargaining power—has received far less attention. Couples disagree systematically about how many children to have, and women across countries report lower preferred fertility than their husbands (Westoff, 2010; Doepke and Tertilt, 2018; Doepke and Kindermann, 2019; Doepke et al., 2023). Whichever spouse holds more bargaining power tilts realized fertility toward their own preference, so any technology that shifts the relative earnings of husbands and wives moves fertility through bargaining, not just through opportunity-cost and income. This mechanism is especially salient in developing countries, where women’s bargaining position remains constrained by entrenched gender norms, unequal property and inheritance rights, and persistent labor-market discrimination, so that realized fertility systematically diverges from women’s own preferred level (Duflo, 2012; Ashraf et al., 2014). The three forces need not pull in the same direction; the net effect is theoretically ambiguous, and which channel dominates is the empirical question of interest.

The latest wave of automation is the natural test case. Industrial robots have spread rapidly across the world economy and generated striking labor-market substitutions, with the 2016 World Development Report estimating that roughly two-thirds of jobs in developing countries—about 1.8 billion positions—are susceptible to automation. The displacement is not gender-neutral: robot adoption is concentrated in male-dominated manufacturing sectors, disproportionately displaces male workers, and narrows gender gaps in both employment and earnings (Acemoglu and Restrepo, 2020; Lerch, 2025; Giuntella et al., 2025). Over the same period, fertility has fallen sharply across

both developed and developing economies. In this paper, we test whether these trends are causally linked, and through which of the income, opportunity-cost, and bargaining channels a male-biased technology shock transmits to fertility, using evidence from China.

China provides the ideal institutional setting: First, fertility is the policy puzzle. Despite the repeated relaxations of birth quotas in recent years, the Chinese total fertility rate has fallen to approximately 1.0—among the lowest ever recorded for a major economy—making the persistent decline a first-order puzzle for both macro and family economics, and the policy circles.¹ Second, China is the dominant market for robots. It has been the world’s largest installer of industrial robots since 2013, accounting for more than half of global new installations by the end of our sample period, with the bulk of new installations concentrated in male-dominated manufacturing sectors that supply the gender-asymmetric labor market shock at the heart of the analysis (Cheng et al., 2019; Deng et al., 2025; Giuntella et al., 2025). The scale of adoption also produces substantial cross-prefecture variation in exposure to robots. Third, women’s initial bargaining position varies substantially across regions and households—partially constrained by social norms—providing the cross-sectional variation needed to highlight the importance of bargaining channel.

To discipline the analysis, we develop a household bargaining model in which fertility is jointly determined by spouses with heterogeneous preferences. Children are produced using complementary time inputs from both spouses, and fertility choices emerge from a Pareto-efficient bargaining process; the husband’s bargaining weight rises with his relative wage (Chiappori, 1988, 1992); and the wife’s effective preference for fertility is depressed by the share of an unfavorable default child-care burden she anticipates carrying (Lundberg and Pollak, 1993; Doepke and Kindermann, 2019). We study a male-biased technology shock that disproportionately displaces male workers, raising women’s relative earnings within the household. The model decomposes the fertility response into three channels: a negative income effect (lower household resources reduce the affordability of children), a positive opportunity-cost effect (lower wages make childcare time cheaper), and a negative bargaining effect (the shock shifts power toward wives, who systematically desire fewer children than their husbands). The net effect is theoretically ambiguous. Importantly, the model

¹China’s total fertility rate fell to approximately 1.0 in 2023, among the lowest ever recorded for a major economy. See World Bank, World Development Indicators, available at <https://genderdata.worldbank.org/en/indicator/sp-dyn-tfrr-in>; OECD Family Database, available at <https://data.oecd.org/pop/fertility-rates.htm>.

also delivers two testable predictions that isolate the bargaining channel from the classic income and opportunity-cost channels. First, the fertility decline is larger where the spousal preference gap is wider, because the shock shifts decision-making power toward wives, who desire fewer children than their husbands; the wider the gap, the more the realized outcome moves toward the wife’s lower preferred fertility. Second, the decline is more pronounced in regions with traditional gender norms, because a more unfavorable default childcare burden makes a woman’s bargaining position more sensitive to wage shifts.

We test the model channels and predictions using the rapid diffusion of industrial robots across regions in China. We identify the causal effect using a Bartik shift-share instrument that measures prefecture-level robot exposure by combining China’s pre-determined industrial employment structure with exogenous variation in robot adoption across U.S. industries—a strategy that isolates global technological push from local demographic pull (Acemoglu and Restrepo, 2020). Our baseline estimates show that robot exposure significantly *reduces* fertility during 2010-2018: a one-standard-deviation increase in exposure lowers the number of children by 3 percent relative to the sample mean, an effect that operates primarily along the intensive margin (fewer children among parents) rather than the extensive margin (entry into parenthood). This estimate is corroborated by prefecture-level census data, confirming that the pattern generalizes beyond the survey sample. Our finding is also robust to extensive sensitivity checks: validation of both the shift and share components of the instrument (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022), stacked first-difference specifications, controls for major fertility and labor-market policies (the relaxation of the One-Child Policy, maternity leave extensions, and minimum wage changes), province-year fixed effects, prefecture-specific time trends, alternative instruments, and alternative clustering structures.

We then show that robot adoption constitutes a male-biased technology shock. Robot exposure displaces workers of both sexes but falls disproportionately on men: a one-standard-deviation increase in exposure reduces the husband’s employment, working hours, and earnings by substantially more than the corresponding losses for the wife. As a result, robot exposure significantly narrows within-household gender gaps, raising women’s earnings relative to their husbands’.

Turning to mechanisms, we find that the evidence strongly supports the bargaining channel.

Greater robot exposure increases both the probability that wives earn more than their husbands and wives' share of household earnings. Using more direct measures of bargaining power, we find that robot exposure also raises the likelihood that wives are the most informed about household finances and that households spend more on female-related items.

To sharpen this interpretation, we exploit the two forces that isolate the bargaining channel: the spousal preference gap in desired fertility and gender norms. Both shift *only* the bargaining channel, leaving the income and opportunity-cost channels unaffected, so heterogeneity along these two dimensions isolates bargaining from the income and opportunity-cost channels. Indeed, the fertility response is sharply amplified where traditional gender norms are stronger and where the spousal gap in desired fertility is wider, and is small and insignificant elsewhere.

By contrast, the opportunity-cost channel predicts that lower earnings make childcare cheaper, raising fertility; this channel works against our baseline finding and therefore cannot be the dominant mechanism. We do find evidence that industrial robot exposure reduces household income and thus contributes to the fertility decline. The heterogeneity analysis on preference gap and gender norms indicates, however, that the income channel cannot fully explain that decline.²

Our paper contributes to four strands of literature. First, we add to the growing body of evidence that intra-household bargaining shapes fertility when spouses have heterogeneous preferences (Thomson, 1997; Thomson and Hoem, 1998; Ashraf et al., 2014; Hener, 2015; Doepke and Tertilt, 2018; Hazan et al., 2021): We show that an exogenous macroeconomic shock—robot adoption—can shift that balance durably, with realized fertility moving toward women's systematically lower fertility preferences as women's bargaining power strengthens. This mechanism is consistent with the theoretical frameworks of Rasul (2008) and Eswaran (2002), and with recent work emphasizing the role of persistent gender conflict over fertility (Doepke and Tertilt, 2018; Doepke et al., 2023; Goldin, 2025).

Second, we extend the literature on industrial robots and family outcomes (Anelli et al., 2024; Costanzo, 2025; Gihleb et al., 2026; Giuntella et al., 2025; Yu et al., 2024) by providing the first

²We also examine two alternative explanations: (i) a quality–quantity substitution, which we find no support for—robot exposure has no significant effect on educational spending or children's academic performance; and (ii) postponement rather than permanent reduction in fertility, which the data also disfavor—the negative effect strengthens at longer lags with no compensatory rebound among older cohorts.

mechanism-disaggregated evidence in a developing-country setting. Existing studies from developed economies show that exposure to industrial robots affects family behavior through labor market disruptions, influencing marital stability and fertility timing (Anelli et al., 2024; Costanzo, 2025). Evidence from China also documents a negative effect of robot exposure on fertility (Giuntella et al., 2025; Yu et al., 2024), but these studies mainly emphasize aggregate fertility effect and labor market outcomes without an in-depth examination of the underlying mechanisms. While Gihleb et al. (2026) show that industrial robots alter women’s intra-household bargaining power and discuss this as a potential explanation for the fertility-reducing effects of robot adoption in the U.S., they neither formalize the underlying theoretical mechanism nor empirically test its key assumptions. We advance this literature by developing a formal model that separates the income, opportunity-cost, and bargaining channels, testing its assumptions directly, and—most distinctively—using heterogeneity along dimensions that by construction shift only the bargaining channel (the spousal preference gap and gender norms) to isolate it from the classic income and opportunity-cost channels.

Third, we complement the technology-and-fertility literature—which has emphasized birth control innovations (Goldin and Katz, 2002; Bailey, 2006; Lesthaeghe, 2010; Ananat and Hungerman, 2012; Bailey, 2025) and household production technologies (Greenwood et al., 2005, 2023)—by showing that production-side technologies with gender-asymmetric labor market effects can be a quantitatively important driver of fertility decline.³ Industrial robots are one empirically tractable instance of this broader mechanism, but the logic extends to other gender-biased technologies—including AI-driven automation and trade-induced structural change—that differentially alter male and female labor demand.

Fourth, we also contribute to the literature on China’s fertility puzzle—the persistent failure of fertility to rebound after repeated relaxations of birth quotas. After decades of fertility-suppressing measures (Zhang, 2017; Chen and Fang, 2021), China has staged successive reversals from the One-Child Policy through the 2021 Three-Child Policy, yet aggregate fertility has continued to drift downward (Jones, 2022; Goldin, 2025). Fang and Liu (2026) show that the Universal Two-Child

³Exploiting the diffusion of a gender-biased agricultural technology—genetically engineered soy—in Brazil, Moorthy (2025) shows that it reduced female earnings and employment while increasing fertility through income effects and a decline in the opportunity cost of childbearing. Billari et al. (2019) examine the impact of broadband Internet on fertility and find a positive effect among highly educated women aged 25 and above, as improved connectivity facilitates the reconciliation of career and motherhood.

Policy raised births by only 0.023 per eligible woman—almost entirely among women already desiring two or more children—and argue that women’s *desired* fertility, not administrative quotas, has become the binding constraint, reviving classical demographic insights (Easterlin, 1975; Pritchett, 1994). A parallel literature traces this decline to rising childrearing costs, housing prices, and educational competition in East Asia (Anderson and Kohler, 2013; Meng et al., 2023; Kim et al., 2024; Meng et al., 2025). We contribute a macro-structural driver: gender-biased technological change moves realized fertility downward by reweighting whose preferences bind within the household.

The remainder of the paper is organized as follows. Section 2 develops the theoretical framework. Section 3 provides institutional context. Section 4 outlines the empirical strategy. Section 5 describes the data sources, sample construction and summary statistics. Section 6 presents the baseline fertility results, characterizes the gender-biased shock, and reports a battery of robustness checks. Section 7 examines the model mechanisms and predictions. Section 8 concludes.

2 Theoretical Framework

We develop a household bargaining model to formalize how a gender-biased technology shock affects fertility. The partners bargain over consumption and fertility taking into consideration of the *ex ante* non contractible childcare effort.

The model yields three channels—income, opportunity cost, and bargaining—and delivers a set of empirically testable sufficient conditions under which the net fertility response is negative. The formal derivations are collected in Appendix B.

Households and preferences.

A household consists of a male partner m and a female partner f . Each spouse enjoys private consumption and the number of children n (a household public good):

$$u_m = \ln c_m + \beta_m \ln n, \quad (1)$$

$$u_f = \ln c_f + \beta_f \ln n, \quad (2)$$

where $\beta_m, \beta_f > 0$ capture fertility preferences. We focus on the asymmetric preference case that $\beta_m > \beta_f$ that husbands desire more children than wives. This is consistent with findings across countries (Westoff, 2010; Doepke and Tertilt, 2018; Doepke et al., 2023), as well as in China (see Figures 3 in Section 3 for details).

Producing children requires time inputs from both spouses. Let t_m, t_f denote time devoted to childcare. Children are produced using a constant elasticity of substitution (CES) technology:

$$n = \left[(v_m t_m)^{\frac{\sigma-1}{\sigma}} + (v_f t_f)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (3)$$

where $v_m, v_f > 0$ are efficiency parameters and $0 < \sigma < 1$ is the elasticity of substitution, and male and female time are complements in childcare.

Let w_m, w_f denote market wages. Cost minimization yields a unit cost (price per child) give by $p(w_m, w_f) = \left[\left(\frac{w_m}{v_m} \right)^{1-\sigma} + \left(\frac{w_f}{v_f} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$. (see Appendix B)

Denote $\theta_m \equiv \partial \ln p / \partial \ln w_m$ and $\theta_f \equiv 1 - \theta_m$, to measure how each spouse's wage feeds into the cost of raising children. A wage cut for either spouse cheapens children most when that spouse contributes a large share of childcare time. Under $\sigma < 1$, both shares are interior: $\theta_m, \theta_f \in (0, 1)$.

Let $S \equiv w_m + w_f$ denote the full household income. The household budget constraint is given by $c_m + c_f + p(w_m, w_f)n = S$.

Default childcare burden and effective preferences.

Since childcare effort is not contractible *ex ante*, a wife would account for the *ex post* childcare cost in her valuation of an additional child. Doepke and Kindermann (2019) encode this through a two-stage bargaining problem in which spouses re-bargain consumption and childcare after a birth; we collapse the same logic into a one-stage cooperative program by absorbing the wife's anticipated post-birth loss into an effective fertility preference.

Let $\tau_f \in (0, 1)$ denote the wife's default share of childcare time after children are born. τ_f can come from both the social and institutional environment that shapes the default division of unpaid care work. For example, stronger traditional gender norms—such as the belief that women should prioritize family over career—raise τ_f , while better social support for childcare—publicly subsidized or market-based childcare, generous parental leave policies, or proximity to grandparents—lowers it (Lundberg and Pollak, 1993; Doepke and Kindermann, 2019). A wife who anticipates carrying the bulk of caregiving (higher τ_f) therefore prefers to have less children at the bargaining stage. We capture this commitment friction by an effective female fertility preference⁴

$$\beta_f^{\text{eff}} \equiv \beta_f - \gamma \tau_f, \quad (4)$$

where $\gamma > 0$ measures how strongly an unfavorable default depresses the wife's continuation value. Appendix B microfound (4) as the local equivalent of the explicit two-stage game with post-birth renegotiation in the spirit of Doepke and Kindermann (2019). We assume that the wife bears a

⁴The husband's effective fertility preference could change to $\beta_m^{\text{eff}} = \beta_m - \gamma_m \tau_m = \beta_m - \gamma_m(1 - \tau_f)$ with $\gamma_m > 0$. We set $\gamma_m = 0$ for tractability. It also reflects the asymmetric nature of the commitment friction—the wife is the constrained party who bears the disproportionate default childcare burden—following Doepke and Kindermann (2019).

larger-than-equal default childcare burden, $\tau_f > \frac{1}{2}$. And in the following analysis, we focus on the case with interior effective preference $\beta_f^{\text{eff}} > 0$, equivalently $\gamma\tau_f < \beta_f$.

Fertility choice.

The fertility decision is a cooperative Pareto program with weight $\lambda \in (0, 1)$ on the male partner:

$$U = \lambda \ln c_m + (1 - \lambda) \ln c_f + \bar{\beta} \ln n, \quad \bar{\beta} \equiv \lambda\beta_m + (1 - \lambda)\beta_f^{\text{eff}}. \quad (5)$$

Solving (5) subject to the budget constraint yields the closed-form fertility choice

$$n^* = \frac{\bar{\beta}}{1 + \bar{\beta}} \cdot \frac{S}{p(w_m, w_f)}. \quad (6)$$

Realized fertility scales with full income, falls in the unit cost of children, and rises in the household's composite preference $\bar{\beta}$, which mixes the two spouses' weights through the bargaining weight λ .

Technology shock and endogenous bargaining

Let $\rho \geq 0$ denote exposure to a gender-biased technology shock. Wages respond multiplicatively:

$$w_m(\rho) = w_m^0 \exp(-\delta_m \rho), \quad w_f(\rho) = w_f^0 \exp(-\delta_f \rho), \quad (7)$$

where $\delta_m, \delta_f > 0$ are wage semi-elasticities. In the empirical application, ρ measures local exposure to industrial robots, but the framework applies to any technology that differentially affects male and female wages. We focus on robot adoption as a **male biased technology shock** that $\delta_m > \delta_f > 0$.

Following the collective household model tradition (Chiappori, 1988, 1992), we model the bargaining weight as a reduced-form function of the male-to-female wage ratio $r = w_m/w_f$, the canonical distribution factor in this literature. The wage ratio summarizes the relative value of each spouse's market alternative, and extensive empirical work confirms that it tracks revealed bargain-

ing power within households (Thomas, 1990; Browning et al., 1994; Duflo, 2003).

$$r \equiv \frac{w_m}{w_f}, \quad \lambda = \Lambda(r), \quad \Lambda'(r) > 0. \quad (8)$$

Lemma 1 (Bargaining response). *Under (7) and (8), $d\lambda/d\rho = -\kappa(r)(\delta_m - \delta_f) < 0$, where $\kappa(r) \equiv r\Lambda'(r) > 0$.*

Three channels.

Differentiating $\ln n^*$ from (6) with respect to ρ at fixed τ_f produces a transparent three-way decomposition:

$$\frac{d \ln n^*}{d\rho} = \underbrace{-(\delta_m s_m + \delta_f s_f)}_{\text{Income } (<0)} + \underbrace{(\theta_m \delta_m + \theta_f \delta_f)}_{\text{Opportunity cost } (>0)} - \underbrace{\frac{\beta_m - \beta_f^{\text{eff}}}{\beta(1 + \beta)} \kappa(r)(\delta_m - \delta_f)}_{\text{Bargaining } (<0)}. \quad (9)$$

- the *income channel*, which is negative because the shock contracts household earnings;
- the *opportunity-cost channel*, which is positive because lower wages cheapen child-rearing time;
- the *bargaining channel*, which is negative because the shock shifts weight toward the lower-fertility-preference spouse.

The first two are classic channels emphasized in the literature, while the third is less explored. The total response is therefore theoretically ambiguous.

Beyond the sign of the total effect, the decomposition delivers sharp comparative statics. We first state two results that *isolate* the bargaining channel.

Proposition 1 (Norm amplification). *A higher default childcare burden amplifies the bargaining channel, and therefore the fertility-reducing effect of the technology shock.*

$$\frac{\partial^2 \ln n^*}{\partial \rho \partial \tau_f} < 0.$$

Proposition 2 (Preference-gap amplification). *A wider spousal gap in fertility preferences amplifies the bargaining channel, and therefore the fertility-reducing effect of the technology shock:*

$$\frac{\partial^2 \ln n^*}{\partial \rho \partial (\beta_m - \beta_f)} < 0.$$

A higher default share τ_f or a wider primitive preference gap $\beta_m - \beta_f$ both widen the effective preference gap $\beta_m - \beta_f^{\text{eff}}$ and lower the household composite $\bar{\beta}(1 + \bar{\beta})$, making the bargaining channel respond more sensitively to the same wage shock. The economic content is that traditional gender norms and spousal disagreement over family size do not create the bargaining channel; they amplify its transmission. In norm-bound households, or where spouses disagree more about how many children to have, a given shift in bargaining power translates into a larger movement in realized fertility. Appendix B provides the proofs.

Remark 1 (Isolation of the bargaining channel). *The income and opportunity-cost terms in (9) contain no preference parameters: they depend only on wage semi-elasticities and the income and childcare-cost shares. Both therefore drop out of the cross-partials in Propositions 1 and 2. Heterogeneity in the fertility response with respect to τ_f and to $\beta_m - \beta_f$ thus identifies the bargaining channel alone: neither the income nor the opportunity-cost channel predicts any such variation.*

3 Institutional Background

We take the model to the data using evidence from China. This section describes the institutional setting that shapes both the fertility margin, robot adoption and the regional variation in gender norms we exploit.

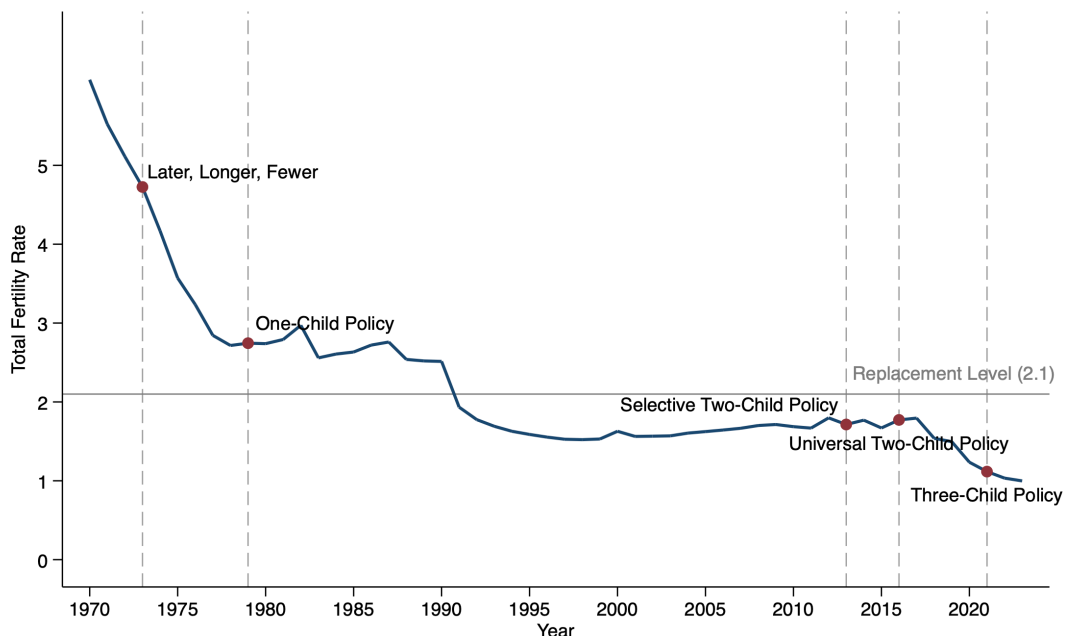
Fertility policy and the fertility puzzle.

China’s fertility trajectory has been shaped by decades of explicit population planning. The “later, longer, and fewer” campaign, launched in 1973, promoted later marriage and childbearing, longer birth intervals, and fewer children. It was followed by the One-Child Policy, introduced in 1979, which imposed stricter limits on family size through financial penalties, employment sanctions, and administrative enforcement, with urban couples generally restricted to a single child and rural families permitted a second birth if the first was a daughter (Hesketh et al., 2005; Zhang, 2017). The policy succeeded in sharply reducing fertility, but by the 2010s policymakers grew concerned about rapid population aging and began a staged relaxation: the Selective Two-Child Policy (2013) allowed couples in which at least one spouse was an only child to have a second birth; the Universal Two-Child Policy (2016) extended this permission to all couples; and the Three-Child Policy (2021) further raised the ceiling to three children per couple (Fang and Liu, 2026). Yet fertility has not rebounded. China’s total fertility rate fell to approximately 1.0 in 2023—among the lowest ever recorded for a major economy—raising first-order concerns about labor supply, pension sustainability, and long-run growth (Jones, 2022; Goldin, 2025). See Figure 1 for the evolution of fertility in China. Understanding what continues to drive fertility downward in the face of pro-natalist policy is therefore a central question for both research and policy.

Rapid robot adoption.

Alongside this demographic decline, China has experienced a rapid expansion in the adoption of industrial robots. Since 2013, it has been the world’s largest market, with diffusion accelerating across manufacturing and related sectors.

Figure 1: China’s Total Fertility Rate and Major Fertility Policy Changes, 1970–2023



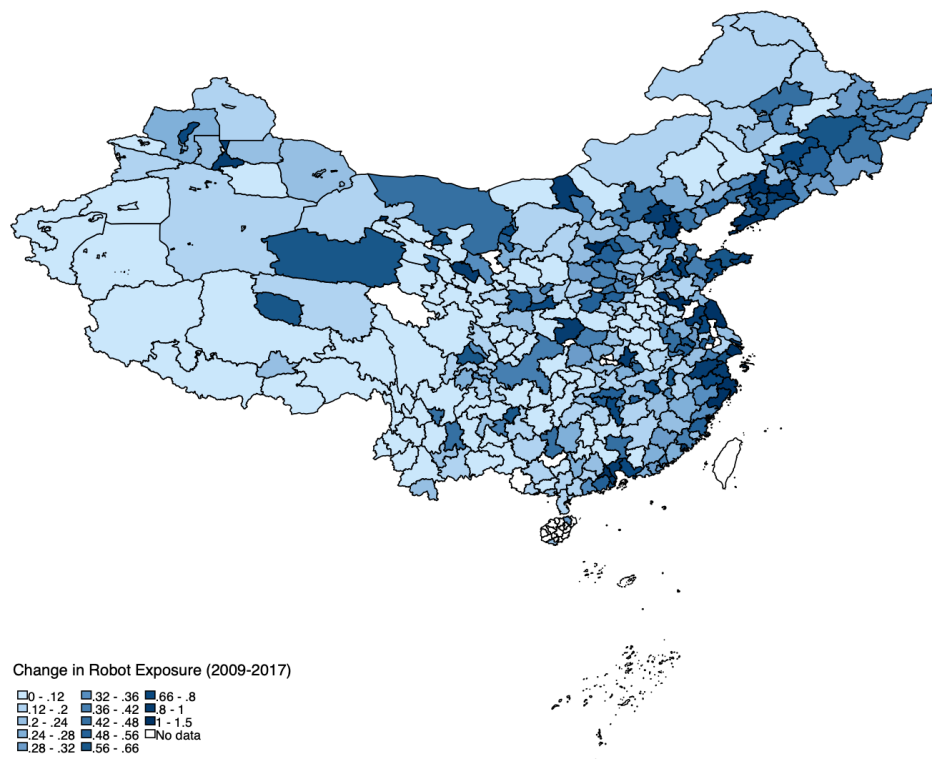
Notes: The figure plots China’s total fertility rate from 1970 to 2023, using data from the World Bank. Vertical dashed lines mark selected policy milestones: “Later, Longer, Fewer” (1973), One-Child Policy (1979), Selective Two-Child Policy (2013), Universal Two-Child Policy (2016), and Three-Child Policy (2021). The horizontal gray solid line at 2.1 indicates the replacement level of fertility.

Robot adoption is also highly uneven across space. As Figure 2 shows, while the increase between 2009 and 2017 is concentrated in eastern coastal prefectures, many inland and southwestern prefectures with long-established manufacturing bases also experience large exposures. This largely reflects pre-existing industrial structure: prefectures with larger pre-2000 employment shares in automotive, electronics, electrical machinery, and metal products—the sectors most amenable to robotic automation—absorbed disproportionately more of the post-2009 robot stock. The resulting cross-prefecture variation is the identifying variation that the Bartik shift-share instrument in Section 4 exploits, and it was largely determined before the period under study.

Gender norms and fertility preferences.

China has a long tradition of gender norms assigning men to work and women to domestic roles (“men are breadwinners, women are homemakers”). Rooted in Confucian ideology (e.g., “male superiority and female inferiority”), these norms were historically reinforced by formal and informal institutions such as foot-binding, widow chastity, patrilineal inheritance rules, and the

Figure 2: Change in Robot Exposure Across Chinese Prefectures, 2009–2017



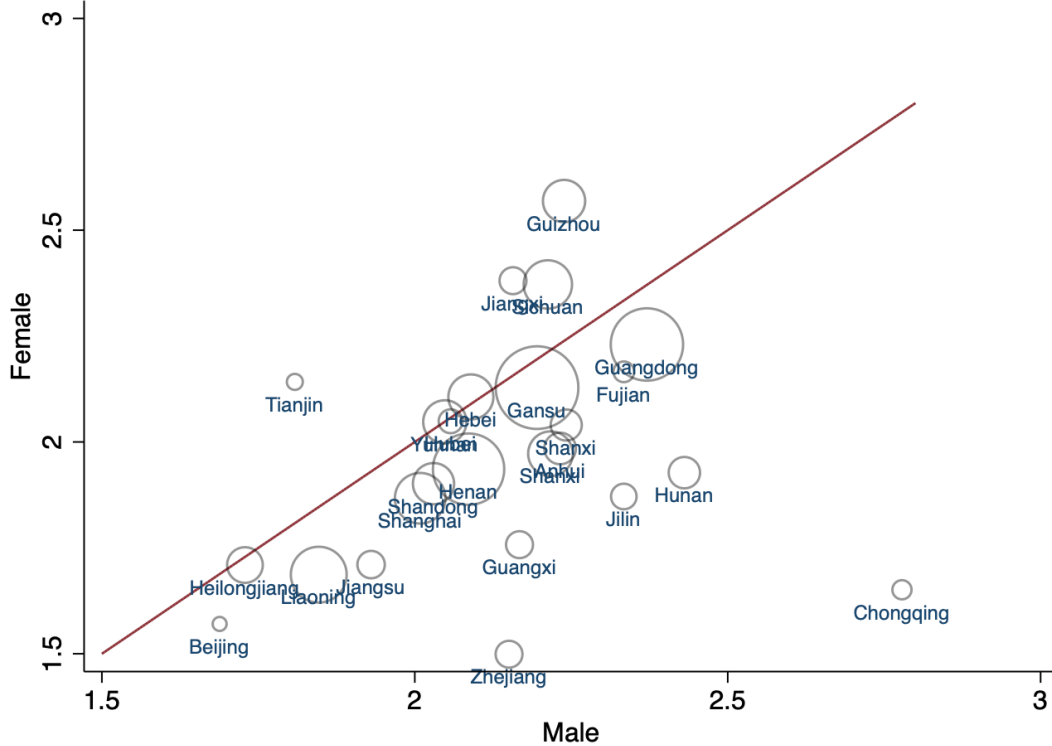
Notes: The figure maps the prefecture-level change in industrial robot exposure between 2009 and 2017. Robot exposure is computed using Equation (11). Darker blue indicates a larger increase in robot exposure. The sample covers all Chinese prefectures.

keju civil examination system (Hinsch, 1994; Fan and Wu, 2023). Although the founding of the People’s Republic of China in 1949 and the subsequent planned economy temporarily weakened them, the traditional division of labor was never permanently displaced (Entwisle and Henderson, 2000; Li, 2021); despite later legal progress, strong intergenerational transmission has kept these norms persistent, with substantial variation in their intensity across regions.

As a result, women in China hold relatively weaker intra-household bargaining power and bear a disproportionate share of unpaid domestic labor (Chen and Hu, 2021): CFPS data show that wives spend 2.234 hours per day on housework—more than twice the 1.012 hours spent by husbands—accounting for 70.2% of total household labor. This imbalance raises the relative cost of childbearing for women (Meng et al., 2023; Zhang et al., 2024), amplifying a gender gap in fertility preferences that is itself near-universal: women across countries report systematically lower desired fertility than men (Westoff, 2010; Doepke and Tertilt, 2018; Doepke and Kindermann, 2019; Doepke et al., 2023). Consistent with this, CFPS evidence shows that unmarried Chinese women report a mean

desired fertility of 1.66, compared with 1.72 for men. The gap is pervasive across provinces—nearly every province lies below the 45-degree line of equal preferences—but its magnitude varies substantially, partly reflecting cross-regional differences in the intensity of traditional gender norms (Figure 3). We also use these variations to further isolate the bargaining channel from the income and opportunity cost channels in mechanism tests.

Figure 3: Gender Differences in Desired Number of Children across Provinces



Notes: This figure illustrates the gender difference in the average desired number of children across provinces. The data are drawn from the 2014 and 2018 waves of CFPS. The sample is restricted to unmarried men and women to mitigate potential endogeneity concerns arising from the possibility that desired fertility may be influenced by a spouse after marriage. The vertical axis shows the average desired number of children among females at the provincial level, while the horizontal axis shows the corresponding average for males. The size of the bubble represents the province’s sample size.

Taken together, the fertility puzzle, rapid robot adoption, and the persistence of traditional gender norms, together with the resulting gender gap in fertility preferences, make China a particularly suitable setting to examine whether gender-biased technological change affects fertility through intra-household bargaining.

4 Empirical Strategy

We study the effects of industrial robot exposure on labor market outcomes, fertility, and related household behaviors using the following specification:

$$Y_{ict} = \alpha + \beta \text{RobotExposure}_{c,t-1} + \mathbf{X}_{ict}\gamma + \delta_i + \xi_t + \varepsilon_{ict}. \quad (10)$$

where subscripts i , c , and t denote individuals, prefectures, and years (or survey waves), respectively. Y_{ict} denotes the outcome variable for individual i in year t , encompassing fertility outcomes (number of children and an indicator for having any children), labor market outcomes, and related household behaviors. The key explanatory variable, $\text{RobotExposure}_{c,t-1}$, measures exposure to industrial robots in prefecture c in year $t - 1$, constructed following Equation (11). $\mathbf{X}_{ict}$ is a vector of control variables, δ_i and ξ_t denote individual and year fixed effects, respectively, and ε_{ict} is the error term. Standard errors are clustered at the prefecture level.

The vector of controls $\mathbf{X}_{ict}$ includes three sets of variables. The first set consists of individual and spousal demographics, including the respondent’s age, age squared, years of education, hukou status,⁵ as well as the spouse’s age, age squared, and years of education. The second set interacts baseline (2010) prefecture characteristics with a linear time trend, $t \times \mathbf{Z}_{c,2010}$, to absorb time-varying local confounders correlated with both robot exposure and fertility. These characteristics include the natural population growth rate, the tertiary-sector employment share, total employment, and GDP. The third set includes province-specific linear time trends, $t \times \sigma_p$, which absorb smoothly evolving province-level confounders—such as long-run demographic transitions, economic restructuring, and slow-moving cultural or institutional changes—that may jointly affect robot adoption and fertility.

Following [Acemoglu and Restrepo \(2020\)](#), $\text{RobotExposure}_{ct}$ is constructed using a shift-share approach that combines cross-prefecture variation in industrial employment structures in 2000 with nationwide changes in robot adoption across industries. Formally, robot exposure is defined as:

$$\text{RobotExposure}_{ct} = \sum_k \left(\frac{L_{ck,2000}}{\sum_k L_{ck,2000}} \right) \frac{R_{kt}}{L_{k,2000}}, \quad (11)$$

⁵Hukou is China’s household registration system.

where the summation runs over all industrial sectors k covered in the IFR data. R_{kt} denotes the operational stock of industrial robots in sector k and year t in China. $L_{k,2000}$ is total national employment in sector k in 2000, a period preceding the rapid diffusion of industrial robots. $L_{ck,2000}$ represents employment in sector k in prefecture c , computed using data from China’s 2000 Population Census. The term $L_{ck,2000}/\sum_k L_{ck,2000}$ captures prefecture c ’s pre-determined exposure to the robot shock in sector k .

However, the estimates obtained from Equation (10) may be biased for two main reasons. First, reverse causality may arise if fertility behavior and local demographic dynamics influence firms’ incentives to adopt robots (Acemoglu and Restrepo, 2022). Second, omitted variable bias may occur if unobserved prefecture–year shocks simultaneously affect robot deployment and fertility outcomes. To address these endogeneity concerns, we adopt an instrumental variable (IV) strategy.

Following Acemoglu and Restrepo (2020), Giuntella et al. (2025), Gai et al. (2025), and Costanzo (2025), we construct a Bartik-style instrument that combines industry-level robot adoption in the United States with China’s prefecture-level industry employment shares in 2000. Robot deployment in the United States reflects exogenous technological progress at the global robotics frontier and is plausibly unrelated to local fertility trends in China, while pre-determined employment shares capture prefectures’ differential exposure to these global shocks. This strategy allows us to isolate variation in industrial robot exposure that is driven by external technological advances rather than local demographic or economic conditions. Specifically, the instrument is defined as:

$$RobotExposure_{ct}^{IV} = \sum_{k \in K} \left(\frac{L_{ck,2000}}{\sum_k L_{ck,2000}} \right) \frac{R_{kt}^{US}}{L_{k,2000}^{US}}, \quad (12)$$

where the summation runs over all industrial sectors. R_{kt}^{US} denotes the operational stock of robots in sector k and year t in the United States. $L_{k,2000}^{US}$ is employment in sector k in the United States in 2000. $L_{ck,2000}$ denotes employment in sector k in prefecture c in China. The term $L_{ck,2000}/\sum_k L_{ck,2000}$ therefore represents sector k ’s employment share in prefecture c in the base year.

5 Data and Summary Statistics

5.1 Data Sources

We draw on three data sources. First, industry-level data on the operational stock of industrial robots come from the International Federation of Robotics (IFR), which reports country–industry–year robot counts. Second, individual- and household-level data come from the China Family Panel Studies (CFPS), a nationally representative longitudinal survey conducted biennially since 2010 that covers economic activities, educational attainment, and family relationships. The CFPS is particularly well-suited for our analysis because it tracks both spouses’ labor market outcomes and fertility over time, allowing us to examine intra-household dynamics. To avoid potential confounding from the COVID–19 pandemic, we use the first five waves (2010–2018). Third, we use China’s 2010 and 2020 population censuses to construct prefecture-level total fertility rates, providing near-universal population-level evidence that complements the CFPS panel.

We restrict the main sample from CFPS to non-single females aged 17 to 49, including those who are married or cohabiting and observed in at least two survey waves. This restriction reflects several considerations. First, the theoretical model applies to partnered households. Second, fertility decisions are concentrated among women of reproductive age. Third, out-of-wedlock childbearing remains negligible in China (Myong et al., 2021). We further require that both spouses are observed in the survey data, enabling us to study intra-household decision-making dynamics. We also exclude individuals who are currently enrolled in school, self-employed or primarily engaged in agricultural work, as they are less likely to be affected by industrial robots and their household income is largely non-divisible.⁶ Our final CFPS sample consists of 10,780 observations.

5.2 Summary Statistics

Before turning to the results, we briefly describe the estimation sample. Table A1 (reported in Appendix A) summarizes the main variables. On average, women in the sample have 1.37 children, and 93.9% have at least one child. The mean age of women is 36, with an average of 9 years of

⁶We include these samples in the robustness check.

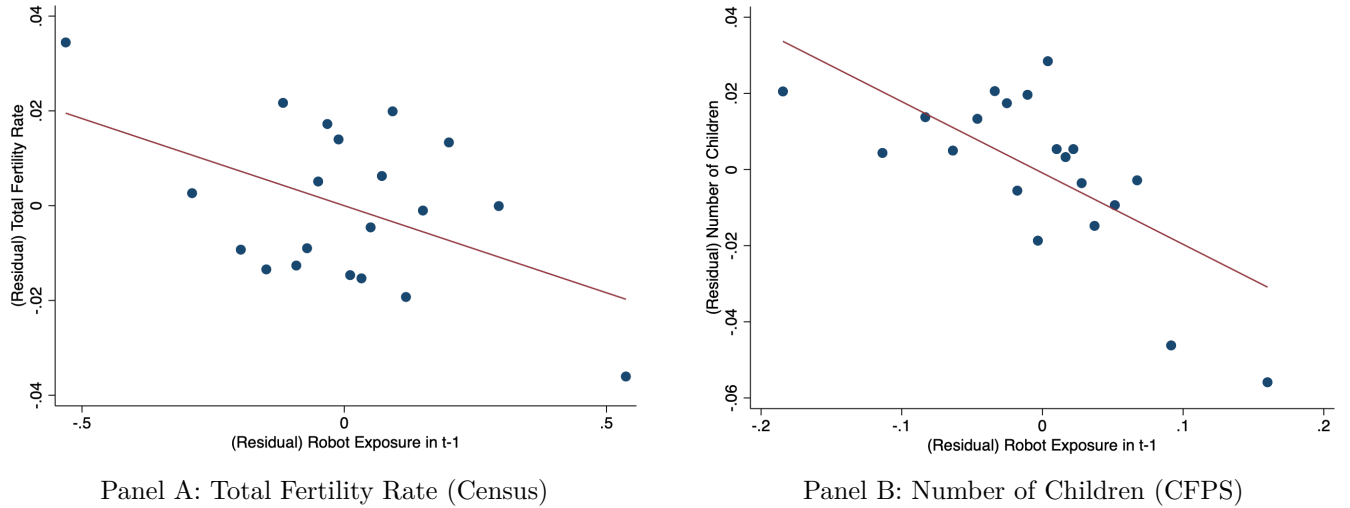
education, and 57.2% hold rural hukou status. Their spouses are slightly older, with an average age of 38 and 10 years of education. In terms of labor market outcomes, 58.5% of women are employed, compared with 87.5% of their spouses. Women work an average of 126 hours per month, whereas their spouses work 187 hours. Annual earnings average 13,740 yuan for women and 25,957 yuan for their spouses. At the household level, total household income—defined as the sum of the woman’s and her spouse’s earnings—averages 39,697 yuan. Regarding women’s economic contribution within the household, 32.8% of women earn income equal to or higher than their spouses, and women contribute, on average, 30.5% of total household income.

6 Baseline Results

6.1 Industrial Robots and Fertility

We first provide a preliminary look at the relationship between robot exposure and fertility graphically in Figure 4. Panel A plots prefecture-level total fertility rates from the 2010 and 2020 population censuses—covering the near-universe of Chinese prefectures—against residual robot exposure. A clear negative gradient emerges: prefectures with greater robot exposure have significantly lower fertility. Panel B confirms this pattern using the CFPS, a longitudinal household survey that enables us to control for individual fixed effects and detailed household characteristics. The consistency across a population-level census and a panel survey motivates the instrumental-variable analysis that follows.

Figure 4: Binned Scatter Plot of Robot Exposure and Fertility



Notes: This figure shows the relationship between residual lagged robot exposure ($t-1$) and residual fertility measures, after controlling for fixed effects. Panel A plots the residual prefecture-level total fertility rate against residual robot exposure, controlling for prefecture fixed effects and province-by-year fixed effects; the data are drawn from the 2010 and 2020 population censuses, which cover the near-universe of Chinese prefectures. Panel B plots the residual number of children against residual robot exposure, controlling for individual fixed effects and province-by-year fixed effects; the data come from the CFPS (2010–2018), with the sample restricted to non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data and for whom neither their own nor their partners’ primary occupations are in agriculture or self-employment.

We now examine how industrial robot exposure causally affects fertility using CFPS.⁷ Table 1

⁷Causal results using Census data are consistent and provided in the robustness check.

reports the baseline results from the 2SLS regressions, with the number of children as the dependent variable. In Column (1), we include year fixed effects and individual fixed effects. To address potential confounding from individual-level time-varying characteristics, Column (2) adds controls for the individual’s age, age squared, years of education, and hukou status, as well as the spouse’s age, age squared, and years of education. To further account for the possibility that prefectural and provincial factors jointly influence robot adoption and fertility, Columns (3) and (4) sequentially add interactions between 2010 prefectural characteristics and a linear time trend, and province-specific linear time trends.

Table 1: Effects of Robot Exposure on Fertility

	Number of Children (Baseline)				Child Dummy	No. Children (≥ 1 child)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Second Stage						
Robot Exposure in $t - 1$	-0.284*** (0.060)	-0.208*** (0.042)	-0.206*** (0.045)	-0.189*** (0.051)	0.065 (0.041)	-0.234*** (0.044)
Panel B. First Stage						
IV	0.136*** (0.008)	0.136*** (0.008)	0.145*** (0.008)	0.161*** (0.008)	0.161*** (0.008)	0.161*** (0.008)
First-stage F Stat.	388.59	387.63	383.82	391.87	391.87	369.79
Year FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y
Individual Controls	N	Y	Y	Y	Y	Y
Prefecture Controls $\times$ Year Trend	N	N	Y	Y	Y	Y
Province FE $\times$ Year Trend	N	N	N	Y	Y	Y
Observations	10,780	10,780	10,780	10,780	10,780	9,938

Notes: This table presents 2SLS estimates of the effects of robot exposure on fertility. The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners’ primary occupations are in agriculture or self-employment. Columns (1)–(4) use the number of children as the dependent variable with progressively richer controls. Column (5) uses an indicator for having at least one child (extensive margin). Column (6) restricts the sample to individuals with at least one child and uses the number of children as the dependent variable (intensive margin). Individual-level controls include age, age squared, education years, rural hukou status, spouse’s age, spouse’s age squared, and spouse’s education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

All first-stage coefficients are positive and significant, with F -statistics far above conventional thresholds, ruling out concerns about weak instruments. The IV estimates on robot exposure are uniformly negative and statistically significant, confirming that industrial robot exposure reduces

fertility. The effect is also economically meaningful: taking the Column (4) estimate as the baseline, a one-standard-deviation increase in robot exposure lowers the number of children by about 3% of the sample mean ($-0.189 \times 0.215/1.37 \approx -0.030$), where 0.215 is the standard deviation of robot exposure in $t - 1$ and 1.37 is the sample mean number of children (both reported in Table A1).

We next decompose the fertility effect into the extensive margin—the probability of having at least one child—and the intensive margin—the number of children among existing parents. Columns (5) and (6) of Table 1 report the results. On the extensive margin (Column (5)), the coefficient on robot exposure is small and statistically insignificant, indicating no detectable effect on entry into parenthood. By contrast, on the intensive-margin sample of parents (Column (6)), the coefficient is negative and statistically significant (-0.234 , $p < 0.01$). The fertility-reducing effect therefore operates primarily along the intensive margin: robot exposure reduces family size among existing parents rather than deterring entry into parenthood.

Robustness

We subject the baseline fertility estimate to an extensive set of robustness checks; all supporting tables are reported in Appendix A.

Cross-validation with census data. A natural concern is whether the fertility effect documented in the CFPS household survey generalizes beyond the survey sample. We replicate the baseline analysis using prefecture-level total fertility rates from China’s population census (Table A2). The census estimates corroborate the survey findings: greater robot exposure is associated with significantly lower TFR, confirming that the negative fertility effect is not an artifact of the CFPS sample or its measurement of fertility.

Instrument validation. We verify the exogeneity of the Bartik shift-share instrument using the Borusyak et al. (2022) balance test for the shift component and Rotemberg weights for the share component. All checks support instrument validity (Tables A3–A5).

Specification robustness. The negative fertility effect is robust across a range of specification checks: a stacked first-difference model that emphasizes within-prefecture and within-individual changes (Table A6); additional controls for the relaxation of the One-Child Policy, maternity leave

extensions, and minimum wage changes (Tables A7–A9); augmentation with province–year fixed effects and prefecture-specific time trends (Table A10); a test ruling out that the results reflect changes in marriage formation rather than within-marriage fertility decisions (Table A11); and incorporation of agricultural and self-employed workers (Table A12).

Data and instrument alternatives. The results are robust to alternative instrumental variables, including those based on industrial robot adoption in Europe and the United States and those using different industry coverage (Table A13). They also remain unchanged when extending the sample to include the 2020 CFPS wave (Table A14) and when clustering standard errors at the provincial or individual level (Table A15).

6.2 A Gender-Biased Shock: Effects on the Labor Market

Having established that robot exposure reduces fertility, we next characterize the shock itself. The theoretical framework in Section 2 assumes that industrial robots constitute a male-biased technology shock; we now test this assumption directly using the same 2SLS specification.

Columns (1) and (2) of Table 2 show that greater exposure to industrial robots significantly reduces the employment probability of both husbands and wives, although the magnitude of the effect is substantially larger for men. Columns (3) and (4) indicate that robot exposure significantly reduces men’s working hours, whereas the estimated effect for women is negative but statistically insignificant. Columns (5) and (6) further show that the effect on women’s earnings is negative but not statistically significant, while men’s earnings decline significantly and by a considerably larger magnitude.

Overall, these results indicate that industrial robot adoption disproportionately affects men’s labor market outcomes, narrowing gender gaps in employment, hours, and earnings. This gender-biased impact confirms that robot adoption is a male-biased technology shock—the key premise of the decomposition in Equation (9). Section 7 examines how this asymmetric shock transmits to fertility through the income, opportunity-cost, and bargaining channels.

Table 2: Effects of Robot Exposure on Gender Gap in Employment, Working Hours, and Earnings

	Wife's Employment	Husband's Employment	Wife's Working Hours (Log)	Husband's Working Hours (Log)	Wife's Earnings (Log)	Husband's Earnings (Log)
	(1)	(2)	(3)	(4)	(5)	(6)
Robot Exposure in $t - 1$	-0.162* (0.087)	-0.348*** (0.076)	-0.039 (0.098)	-0.251** (0.106)	-0.162 (0.235)	-0.270* (0.161)
Individual Controls	Y	Y	Y	Y	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y	Y
Province FE $\times$ Year Trend	Y	Y	Y	Y	Y	Y
Observations	10,780	10,780	6,310	9,435	6,310	9,435
First-stage F Stat.	391.87	391.87	362.25	302.11	362.25	302.11

Notes: This table reports 2SLS estimates of the impact of robot exposure on gender gaps in employment, monthly working hours, and earnings. The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment. The dependent variables in Columns (1) and (2) are indicators for whether the female respondent and her spouse are employed, respectively. The dependent variables in Columns (3) and (4) are the logarithm of monthly working hours for the female respondent and her spouse, respectively. The dependent variables in Columns (5) and (6) are the logarithm of annual earnings for the female respondent and her spouse, respectively. Individual-level controls include age, age squared, education years, rural hukou status, spouse's age, spouse's age squared, and spouse's education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7 Mechanisms and Predictions

Having established the causal effect of industrial robot exposure on fertility and confirmed that the shock is male-biased (Section 6.2), we examine the channels through which it operates. The theoretical framework identifies three forces in Equation (9): an income effect, an opportunity-cost effect, and a bargaining effect. We focus on the bargaining channel, which the model and the data suggest is the dominant mechanism. We first document its direct effect on measured bargaining power, then exploit model-implied heterogeneity to isolate it from the classic channels, and finally discuss the income and opportunity-cost channels.

7.1 The Bargaining Channel

7.1.1 Direct Effect

We begin with the direct effect of robot exposure on intra-household bargaining power. As formalized in the theoretical framework, fertility decisions are jointly determined by both spouses. Consistent with a large body of existing studies (Westoff, 2010; Doepke and Tertilt, 2018; Doepke and Kindermann, 2019) and as illustrated in Figures 3 for China, women typically prefer fewer children than men, in line with the assumption that $\beta_m > \beta_f$. Holding spouses' fertility preferences fixed, a shift in bargaining power toward women would move realized fertility closer to women's desired fertility, thereby lowering fertility on average (Doepke and Tertilt, 2018).

Industrial robots affect male and female labor market outcomes asymmetrically, as documented in Section 6.2, and thus have the potential to alter relative bargaining positions within the household. To examine this mechanism, we construct five proxies for wives' intra-household bargaining power. We begin with two widely used measures based on relative earnings: (1) an indicator for whether the wife earns more than her husband, and (2) the wife's share of total household earnings. These measures provide transparent proxies for bargaining power and are standard in the literature.

We then complement these with two additional proxies that more directly capture wives' intra-household control. First, we include an indicator for whether the wife is the most informed about household finances. Control over financial information is closely linked to decision-making authority

and has been shown to reflect bargaining power over resource allocation (Ashraf, 2009; Bobonis, 2009). Second, we use an indicator for whether women’s consumption share exceeds the sample median. In particular, as the total women’s consumption share is not observable, we use the beauty-related household expenditure as a proxy for women’s consumption (Gihleb et al., 2026). Following the literature, expenditures on private or assignable goods—especially those more closely associated with one spouse—are informative about the intra-household distribution of resources, and thus about bargaining power (Browning et al., 1994; Duflo, 2003; Wang, 2014).

Finally, we construct an aggregate bargaining index following Anderson (2008) by combining the three binary indicators—relative earnings, financial knowledge, and beauty-related expenditure—and standardizing it to have mean zero and unit variance. This composite measure captures multiple dimensions of bargaining power and mitigates measurement error in any single proxy.

Table 3 reports the results. Columns (1) and (2) show that greater exposure to robots increases both the probability that the wife earns more than her husband and her share of household earnings. Columns (3) and (4) show that robot exposure similarly raises the likelihood that the wife is most informed about household finances and that beauty-related household spending exceeds the median. Column (5) confirms that the aggregate index rises with robot exposure, collectively suggesting that wives’ bargaining power within the household is strengthened by robot exposure.

Taken together, the evidence points to intra-household bargaining as the primary channel through which exposure to industrial robots reduces fertility. By asymmetrically reshaping labor market outcomes, robot exposure shifts bargaining power toward wives, so that realized fertility increasingly reflects women’s lower fertility preferences.

7.1.2 Forces Isolating the Bargaining Channel: Preference Gaps and Gender Norms

The evidence shows that robot exposure shifts measured bargaining power toward wives. To confirm that this shift—rather than the income or opportunity-cost channels—drives the fertility decline, we exploit a sharp implication of the model (Remark 1): the spousal preference gap and the strength of gender norms enter *only* the bargaining term of the decomposition in (9). The income and opportunity-cost channels depend on wages and expenditure shares alone and are invariant to these two objects, so any heterogeneity in the fertility response along them isolates the bargaining channel.

Table 3: Effect of Robot Exposure on Intra-household Bargaining Power

	Wife Earns More	Share of Wife's Earnings	Wife Most Informed About Finances	Share of Beauty Expenditure	Aggregate Index
	(1)	(2)	(3)	(4)	(5)
Robot Exposure in $t - 1$	0.526*** (0.108)	0.107** (0.049)	0.149* (0.077)	0.286** (0.115)	0.984*** (0.254)
Individual Controls	Y	Y	Y	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y
Province FE $\times$ Year Trend	Y	Y	Y	Y	Y
Observations	10,780	9,862	10,780	7,768	7,768
First-stage F Stat.	391.87	319.50	391.87	253.05	253.05

Notes: Columns (1)–(3) use data from the CFPS 2010–2018. The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners’ primary occupations are in agriculture or self-employment. Column (1) uses “Wife Earns More” as the dependent variable, a dummy equal to 1 if the wife’s earnings are greater than or equal to the husband’s, and 0 otherwise. Columns (2) uses “Share of Wife’s Earnings” as the dependent variable, defined as the proportion of the wife’s annual earnings in total household income (i.e., Earnings / Household Income). In Column (3), the dependent variable is a dummy indicating whether the wife is the most informed about the household’s financial situation. Column (4) uses data from CFPS 2012–2018. The dependent variable is an indicator equal to one if the share of household expenditure on beauty-related items is above the sample median, and zero otherwise. The dependent variable in Column (5) is an aggregate index constructed from Wife Earns More, Wife Most Informed About Finances, and Share of Beauty Expenditure, following [Anderson \(2008\)](#) (standardized to have mean zero and unit standard deviation). Individual-level controls include age, age squared, education years, rural hukou status, spouse’s age, spouse’s age squared, and spouse’s education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Preference gaps.

Proposition 2 predicts a larger fertility decline where the spousal gap in fertility preferences is wider. We measure the gap at the province level using the average desired number of children reported by *unmarried* men and women. Using unmarried respondents is deliberate: this measure is predetermined before marriage and before shock, and avoids the endogenous revision of stated preferences that can follow marriage and childbearing ([Westoff, 2010](#); [Doepke and Tertilt, 2018](#); [Fang and Liu, 2026](#)). The aggregation at regional level also help alleviate the measurement error

at household level.⁸

Table 4: Heterogeneous Effects by Gender Gap in Desired Number of Children

Gender Gap in Desired Children	Number of Children	
	(1)	(2)
	Large	Small
Robot Exposure in $t - 1$	-0.200** (0.080)	-0.112 (0.091)
Individual Controls	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y
Year FE	Y	Y
Individual FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	5,954	4,826
First-stage F Stat.	124.38	600.07

Notes: This table reports heterogeneous effects of robot exposure on the number of children by the province-level gender gap in desired number of children. To mitigate endogeneity from household-level preference gaps, we construct the gap from the average desired number of children reported by unmarried men and unmarried women at the province level, and split provinces at the median. The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners’ primary occupations are in agriculture or self-employment. Individual-level controls include age, age squared, education years, rural hukou status, spouse’s age, spouse’s age squared, and spouse’s education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

As shown in Table 4, the fertility effect is negative and significant where the gap is large, but statistically insignificant where it is small. Because neither the income nor the opportunity-cost channel varies with the preference gap, this contrast is direct evidence that the bargaining channel drives the decline. Moreover, the absence of an effect in regions with a small preference gap—where the bargaining channel is relatively muted—indicates that the classical income and opportunity-cost channels alone cannot account for the overall fertility decline.

⁸The 2014 and 2018 surveys both asked about desired number of children. However, around 30% of husbands and wives report a different desired number of children across the two waves, reflecting both the endogenous revision of stated preferences and measurement error.

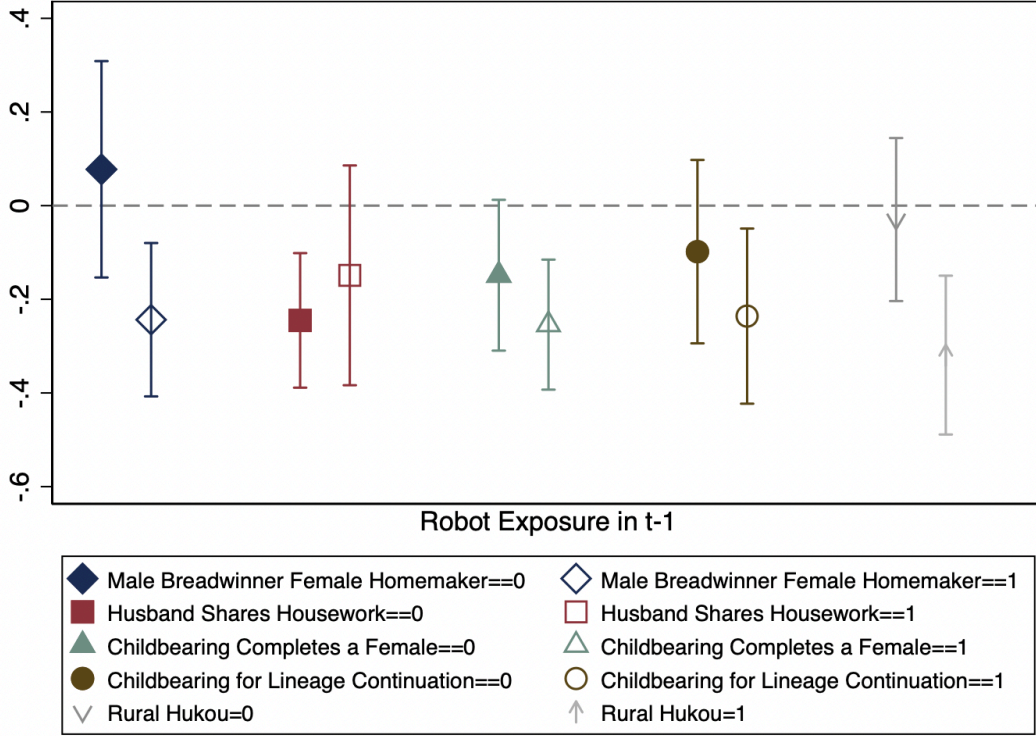
Gender norms.

Proposition 1 predicts that the fertility response to robot exposure is amplified where the default childcare burden τ_f is higher: a larger τ_f widens the effective preference gap and reduces $\bar{\beta}$, both of which magnify the bargaining channel in equation (9). Traditional gender norms are a natural proxy for τ_f , since wives in more norm-bound environments face stronger social expectations to shoulder household and childcare responsibilities.

We test Proposition 1 using five indicators. The first four are prefecture-level survey proxies: (i) *Male Breadwinner Female Homemaker*, based on agreement with the statement “Males should prioritize their careers, while females should prioritize their families”; (ii) *Husband Shares Housework*, based on agreement with the statement “Men should take on at least half of the housework” (the higher the agreement, the weaker the traditional norms); (iii) *Childbearing Completes a Female*, reflecting agreement with “A female is not complete without having children”; and (iv) *Childbearing for Lineage Continuation*, based on agreement with “The purpose of having children is to carry on the family line.” For each attitudinal measure, prefectures with above-median agreement (or below-median for indicator ii) are classified as having stronger traditional norms. The fifth indicator splits the sample by hukou status, since rural residents tend to hold more traditional gender norms and face a higher default childcare burden than their urban counterparts.

Figure 5 plots the estimated coefficient on robot exposure from subgroup regressions for each of the five indicators. The results are strikingly consistent: the fertility-reducing effect is large and statistically significant in the high-norm subgroup across all five measures, while the coefficient is close to zero and statistically insignificant in the low-norm subgroup. The convergence of evidence across attitudinal survey proxies and the hukou split supports Proposition 1.

Figure 5: Heterogeneous Effects by Traditional Gender Norms and Hukou Status



Notes: This figure presents the coefficients on “Robot exposure in $t-1$ ” from subgroup regressions based on alternative measures of traditional gender norms and hukou status. The data come from the CFPS (2010–2018). The regression specification follows Column (4) in Table 1. The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners’ primary occupations are in agriculture or self-employment. “Male Breadwinner Female Homemaker” is based on agreement with the statement “Males should prioritize their careers, while females should prioritize their families”; prefectures with above-median agreement are classified as having stronger traditional norms. “Husband Shares Housework” is based on agreement with the statement that men should take on at least half of the housework; prefectures with above-median agreement are coded as 1 (the higher the agreement, the weaker the traditional norms). “Childbearing Completes a Female” reflects agreement with the statement “A female is not complete without having children.” prefectures with above-median agreement are coded as 1, and 0 otherwise. “Childbearing for Lineage Continuation” is based on agreement with the statement “The purpose of having children is to carry on the family line.” prefectures with above-median support are coded as 1, and 0 otherwise. Error bars represent 95% confidence intervals.

7.2 Income and Opportunity-Cost Channels

Having isolated the bargaining channel, we turn to the two classic channels and show that neither can, by itself, account for the fertility decline. Equation (9) predicts that the opportunity-cost channel *raises* fertility: by reducing wages, robot exposure lowers the time cost of childcare, making children cheaper. The labor market evidence in Section 6.2 shows that women’s earnings decline only modestly, while men’s earnings fall substantially, so the opportunity cost of children should

fall and fertility should increase on this margin alone. Given that we observe a negative fertility response overall, the positive opportunity-cost channel cannot be the dominant mechanism.

The income channel pushes in the opposite direction. Column (1) of Table 5 shows that industrial robot exposure significantly reduces total household income—the combined earnings of both spouses—consistent with the male-biased labor-market effects documented above. The income channel therefore pushes fertility down.

Column (2) adds household income as a control to the fertility regression. The effect of robot on fertility is still significantly negative, suggesting that income effect cannot fully explain the fertility decline. While this specification introduces bad control problem and needs caution to interpret, its result is also consistent with the insignificant heterogeneity results where the bargaining channel is relatively muted in section 7.1.2 (a small preference gap or weak social norm).

Table 5: Effects of Robot Exposure on Household Income and Fertility

	Total Income (Log)	Number of Children
	(1)	(2)
Robot Exposure in $t - 1$	-0.366*** (0.140)	-0.193*** (0.059)
Total Income (Log)		-0.018* (0.010)
Individual Controls	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y
Year FE	Y	Y
Individual FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	9,862	9,862
First-stage F Stat.	272.74	389.31

Notes: The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners’ primary occupations are in agriculture or self-employment. Column (1) uses the logarithm of the sum of annual earnings of the female respondent and her spouse as the dependent variable. Column (2) uses the number of children as the dependent variable and adds total household income (log) as a control. Individual-level controls include age, age squared, education years, rural hukou status, spouse’s age, spouse’s age squared, and spouse’s education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Alternative channels.

We also examine two additional mechanisms. First, industrial robot adoption may influence fertility through a quantity–quality trade-off: by raising the returns to skills, it could increase incentives to invest in child quality rather than quantity (Becker, 1960; Becker and Lewis, 1973; Galor and Weil, 2000; Galor and Moav, 2002; Lagerlöf, 2006; Becker et al., 2010). However, we find no significant effects on educational expenditures or children’s academic performance (Table A16 in Appendix A).

Second, we assess whether the fertility decline reflects a permanent reduction in completed fertility (quantum effect) or a mere postponement (tempo effect) (Costanzo, 2025). Under a tempo mechanism, fertility reductions at younger ages should be offset by higher fertility at older ages, implying little change in completed fertility. By contrast, a quantum effect predicts a persistent decline that becomes more pronounced over longer horizons. Consistent with a quantum effect, evidence from lagged exposure measures and age-specific analyses shows that the negative fertility effect persists and strengthens at longer lags, with no compensatory increase in fertility at older ages (Figures A1–A3 in Appendix A).

8 Conclusion

This paper examines how gender-biased technological change affects fertility through intra-household bargaining, a channel that has received relatively limited attention in the literature. We develop a Pareto-efficient household bargaining model in which spouses have heterogeneous fertility preferences and children require time inputs from both partners. A male-biased technology shock increases women’s relative earnings and bargaining weight, generating three forces on fertility: an income effect, an opportunity-cost effect, and a bargaining effect.

We test the model using the rapid adoption of industrial robots in China. Combining CFPS panel data with a Bartik shift-share instrument, we find that a one-standard-deviation increase in exposure to industrial robots reduces the number of children by approximately 3 percent relative to the mean, operating primarily at the intensive margin. The results are robust to a wide range of specifications.

Mechanism tests point to intra-household bargaining as the dominant mechanism. Specifically, robot adoption disproportionately worsens men’s labor market outcomes, shifts bargaining power toward women, and pulls realized fertility toward women’s systematically lower fertility preferences. We further rule out two alternative interpretations: the decline does not operate through a quantity–quality trade-off, and it reflects a reduction in completed fertility rather than mere postponement. Crucially, heterogeneity in spouses’ fertility preferences is necessary for the bargaining channel to operate; absent such differences, changes in bargaining power would leave fertility unchanged. Because the spousal preference gap and gender norms enter only the bargaining channel, heterogeneity along these dimensions isolates bargaining from the classic channels: the fertility decline is amplified where norms are stronger and where spousal preference gaps are wider, exactly as the model predicts. We further rule out women’s opportunity costs as the primary channel and provide suggestive evidence that income losses alone cannot fully explain the negative fertility effects of industrial robot exposure.

The findings carry clear policy implications. Women’s lower fertility preferences reflect the disproportionate costs they bear—time burdens, career penalties, and unequal childcare responsibilities—imposed by prevailing social and institutional arrangements. Mitigating the fertility

consequences of gender-biased technology, therefore, requires narrowing spousal preference gaps by reducing these cost asymmetries. Promising interventions include expanding affordable, high-quality childcare; strengthening enforcement against pregnancy and motherhood-related workplace discrimination; designing parental leave to preserve women's career continuity while encouraging fathers' uptake; and targeting family support toward more traditional regions and households where preference gaps and bargaining shifts are largest. By addressing the underlying cost asymmetries that depress women's desired fertility, such policies can limit the extent to which shifts in bargaining power induced by gender-biased technology translate into persistent fertility decline.

References

- Acemoglu, D. and P. Restrepo (2020). Robots and jobs: Evidence from U.S. labor markets. *Journal of Political Economy* 128(6), 2188–2244.
- Acemoglu, D. and P. Restrepo (2022). Tasks, automation, and the rise in U.S. wage inequality. *Econometrica* 90(5), 1973–2016.
- Ananat, E. O. and D. M. Hungerman (2012). The power of the pill for the next generation: Oral contraception’s effects on fertility, abortion, and maternal and child characteristics. *Review of Economics and Statistics* 94(1), 37–51.
- Anderson, M. L. (2008). Multiple inference and gender differences in the effects of early intervention: A reevaluation of the abecedarian, perry preschool, and early training projects. *Journal of the American Statistical Association* 103(484), 1481–1495.
- Anderson, T. and H.-P. Kohler (2013). Education fever and the east asian fertility puzzle: A case study of low fertility in South Korea. *Asian Population Studies* 9(2), 196–215.
- Anelli, M., O. Giuntella, and L. Stella (2024). Robots, marriageable men, family, and fertility. *Journal of Human Resources* 59(2), 443–469.
- Ashraf, N. (2009). Spousal control and intra-household decision making: An experimental study in the philippines. *American Economic Review* 99(4), 1245–1277.
- Ashraf, N., E. Field, and J. Lee (2014). Household bargaining and excess fertility: An experimental study in Zambia. *American Economic Review* 104(7), 2210–2237.
- Bailey, M. J. (2006). More power to the pill: The impact of contraceptive freedom on women’s life cycle labor supply. *Quarterly Journal of Economics* 121(1), 289–320.
- Bailey, M. J. (2025). The economics of childbearing: Trends, progress, and challenges. *Annual Review of Economics* 17(1), 695–720.
- Becker, G. S. (1960). An economic analysis of fertility. In *Demographic and Economic Change in Developed Countries*, pp. 209–240. Columbia University Press.
- Becker, G. S. and H. G. Lewis (1973). On the interaction between the quantity and quality of children. *Journal of Political Economy* 81(2), S279–S288.
- Becker, S. O., F. Cinnirella, and L. Woessmann (2010). The trade-off between fertility and education: Evidence from before the demographic transition. *Journal of Economic Growth* 15, 177–204.
- Billari, F. C., O. Giuntella, and L. Stella (2019). Does broadband internet affect fertility? *Population Studies* 73(3), 297–316.
- Bobonis, G. J. (2009). Is the allocation of resources within the household efficient? New evidence from a randomized experiment. *Journal of Political Economy* 117(3), 453–503.
- Borusyak, K., P. Hull, and X. Jaravel (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies* 89(1), 181–213.

- Browning, M., F. Bourguignon, P.-A. Chiappori, and V. Lechene (1994). Income and outcomes: A structural model of intrahousehold allocation. *Journal of Political Economy* 102(6), 1067–1096.
- Chen, Y. and H. Fang (2021). The long-term consequences of China’s “Later, Longer, Fewer” campaign in old age. *Journal of Development Economics* 151, 102664.
- Chen, Y. and D. Hu (2021). Gender norms and marriage satisfaction: Evidence from China. *China Economic Review* 68, 101627.
- Cheng, H., R. Jia, D. Li, and H. Li (2019). The rise of robots in China. *Journal of Economic Perspectives* 33(2), 71–88.
- Chiappori, P.-A. (1988). Rational household labor supply. *Econometrica* 56(1), 63–90.
- Chiappori, P.-A. (1992). Collective labor supply and welfare. *Journal of Political Economy* 100(3), 437–467.
- Costanzo, C. (2025). Robots, jobs, and optimal fertility timing. *Journal of Population Economics* 38, 51.
- Deng, Y., A. Feng, and D. Hu (2025). Gender earnings gap in Chinese firms: Can it be narrowed by industrial robots? *China Economic Review* 89, 102328.
- Doepke, M., A. Hannusch, F. Kindermann, and M. Tertilt (2023). The economics of fertility: A new era. In S. Lundberg, R. A. Pollak, and A. Voena (Eds.), *Handbook of the Economics of the Family*, Volume 1, pp. 151–254. North-Holland.
- Doepke, M. and F. Kindermann (2019). Bargaining over babies: Theory, evidence, and policy implications. *American Economic Review* 109(9), 3264–3306.
- Doepke, M. and M. Tertilt (2018). Women’s empowerment, the gender gap in desired fertility, and fertility outcomes in developing countries. *AEA Papers and Proceedings* 108, 358–362.
- Duflo, E. (2003). Grandmothers and granddaughters: Old-age pensions and intrahousehold allocation in South Africa. *World Bank Economic Review* 17(1), 1–25.
- Duflo, E. (2012). Women empowerment and economic development. *Journal of Economic literature* 50(4), 1051–1079.
- Easterlin, R. A. (1975). An economic framework for fertility analysis. *Studies in Family Planning* 6(3), 54–63.
- Entwisle, B. and G. Henderson (2000). *Re-drawing boundaries: Work, households, and gender in China*, Volume 25. University of California Press.
- Eswaran, M. (2002). The empowerment of women, fertility, and child mortality: Towards a theoretical analysis. *Journal of Population Economics* 15(3), 433–454.
- Fan, X. and L. Wu (2023). The shaping of a gender norm: Marriage, labor, and foot-binding in historical china. *International Economic Review* 64(4), 1819–1850.
- Fang, H. and C. Liu (2026). Desired fertility, realized fertility and the effects of China’s universal two-child policy. *Journal of Development Economics*, 103796.

- Gai, Q., S. Guo, L. Hu, and X. Zhu (2025). Robots, routine jobs, and rural migrant workers: Evidence from China. *Journal of Population Economics* 38(4), 86.
- Galor, O. and O. Moav (2002). Natural selection and the origin of economic growth. *Quarterly Journal of Economics* 117(4), 1133–1191.
- Galor, O. and D. N. Weil (1996). The gender gap, fertility, and growth. *American Economic Review* 86(3), 374–387.
- Galor, O. and D. N. Weil (2000). Population, technology, and growth: From Malthusian stagnation to the demographic transition and beyond. *American Economic Review* 90(4), 806–828.
- Gihleb, R., O. Giuntella, and D. Morag (2026). Gender-specific labour market shocks and household bargaining power. *The Economic Journal*, ueag060.
- Giuntella, O., Y. Lu, and T. Wang (2025). How do workers adjust to robots? Evidence from China. *The Economic Journal* 135(666), 637–652.
- Goldin, C. (2025). Babies and the macroeconomy. *Economica* 92(367), 675–700.
- Goldin, C. and L. F. Katz (2002). The power of the pill: Oral contraceptives and women’s career and marriage decisions. *Journal of Political Economy* 110(4), 730–770.
- Goldsmith-Pinkham, P., I. Sorkin, and H. Swift (2020). Bartik instruments: What, when, why, and how. *American Economic Review* 110(8), 2586–2624.
- Greenwood, J., N. Guner, and R. Marto (2023). The great transition: Kuznets facts for family economists. In J. Greenwood and N. Guner (Eds.), *Handbook of the Economics of the Family*, pp. 389–441. North-Holland.
- Greenwood, J., A. Seshadri, and M. Yorukoglu (2005). Engines of liberation. *Review of Economic Studies* 72(1), 109–133.
- Hazan, M., D. Weiss, and H. Zoabi (2021). Women’s liberation, household revolution. Discussion Paper DP 16838, CEPR.
- Hener, T. (2015). Do couples bargain over fertility? *Annals of Economics and Statistics* 117–118, 211–231.
- Hesketh, T., L. Lu, and Z. W. Xing (2005). The effect of China’s one-child family policy after 25 years. *New England Journal of Medicine* 353(11), 1171–1176.
- Hinsch, B. H. (1994). *Women in early imperial China*. Harvard University.
- Jakobsen, K. M., T. H. Jørgensen, and H. Low (2025). Fertility and family labor supply. Working Paper WP 2025-26, Federal Reserve Bank of Chicago.
- Jones, C. I. (2022). The end of economic growth? Unintended consequences of a declining population. *American Economic Review* 112(11), 3489–3527.
- Jones, L. E., A. Schoonbroodt, and M. Tertilt (2010). Fertility theories: Can they explain the negative fertility–income relationship? In J. B. Shoven (Ed.), *Demography and the Economy*, pp. 43–100. University of Chicago Press.

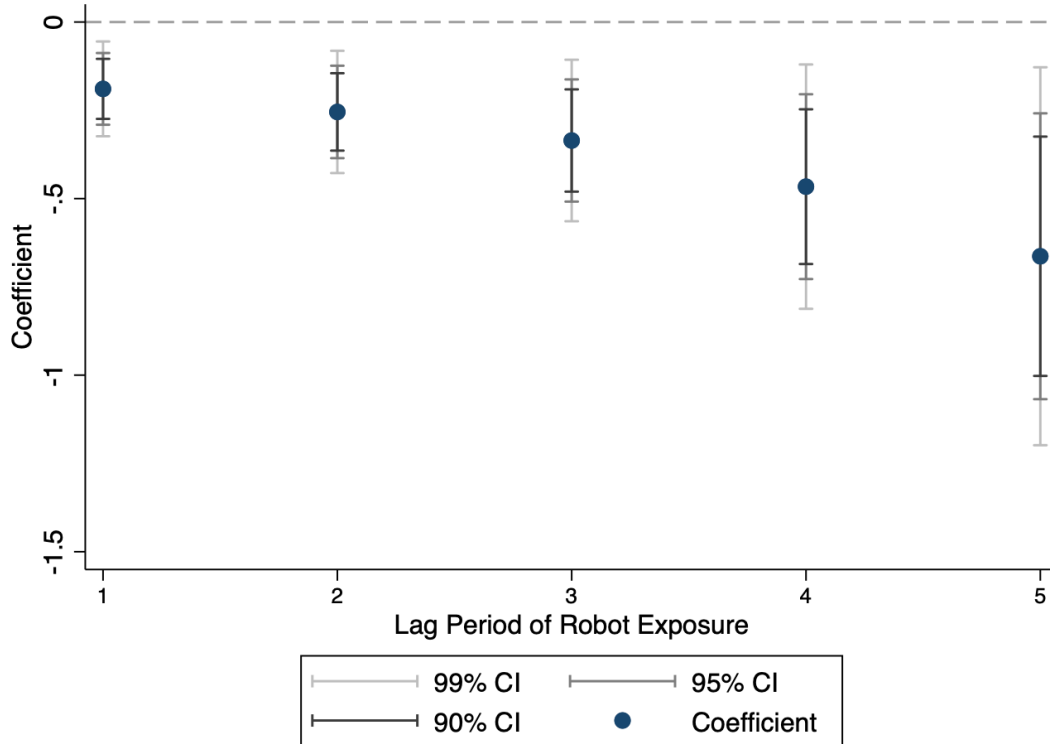
- Kim, S., M. Tertilt, and M. Yum (2024). Status externalities in education and low birth rates in Korea. *American Economic Review* 114(6), 1576–1611.
- Lagerlöf, N.-P. (2006). The Galor–Weil model revisited: A quantitative exercise. *Review of Economic Dynamics* 9(1), 116–142.
- Lerch, B. (2025). From blue- to steel-collar jobs: The decline in employment gaps? *American Economic Journal: Macroeconomics* 17(1), 126–160.
- Lesthaeghe, R. (2010). The unfolding story of the second demographic transition. *Population and Development Review* 36(2), 211–251.
- Li, J. (2021). Women hold up half the sky? trade specialization patterns and work-related gender norms. *Journal of International Economics* 128, 103407.
- Lundberg, S. and R. A. Pollak (1993). Separate spheres bargaining and the marriage market. *Journal of Political Economy* 101(6), 988–1010.
- Meng, J., H. Wang, Y. A. Yang, and M. Zhang (2025). Education competition and fertility intention: Evidence from China’s private tutoring ban. *Economic Journal*, ueaf096.
- Meng, L., L. Peng, and Y. Zhou (2023). Do housing booms reduce fertility intentions? Evidence from the new two-child policy in China. *Regional Science and Urban Economics* 101, 103920.
- Meng, L., Y. Zhang, and B. Zou (2023). The motherhood penalty in china: Magnitudes, trends, and the role of grandparenting. *Journal of Comparative Economics* 51(1), 105–132.
- Mincer, J. (1963). Market prices, opportunity costs, and income effects. In *Measurement in Economics*, pp. 67–82. Stanford University Press.
- Moorthy, V. S. (2025). Agricultural technological change, female earnings and fertility: Evidence from brazil. *The Economic Journal* 135(665), 285–320.
- Myong, S., J. J. Park, and J. Yi (2021). Social norms and fertility. *Journal of the European Economic Association* 19(5), 2429–2466.
- Pritchett, L. H. (1994). Desired fertility and the impact of population policies. *Population and Development Review* 20(1), 1–55.
- Rasul, I. (2008). Household bargaining over fertility: Theory and evidence from Malaysia. *Journal of Development Economics* 86(2), 215–241.
- Schultz, T. P. (1985). Changing world prices, women’s wages, and the fertility transition: Sweden, 1860–1910. *Journal of Political Economy* 93(6), 1126–1154.
- Thomas, D. (1990). Intra-household resource allocation: An inferential approach. *Journal of Human Resources* 25(4), 635–664.
- Thomson, E. (1997). Couple childbearing desires, intentions, and births. *Demography* 34(3), 343–354.
- Thomson, E. and J. M. Hoem (1998). Couple childbearing plans and births in Sweden. *Demography* 35(3), 315–322.

- Wang, S.-Y. (2014). Property rights and intra-household bargaining. *Journal of Development Economics* 107, 192–201.
- Westoff, C. F. (2010). Desired number of children: 2000–2008. DHS Comparative Reports 25, ICF Macro, Calverton.
- Yu, S., J. Zhang, and X. Yan (2024). Robots, gender economic opportunities and household adjustment. *Structural Change and Economic Dynamics* 71, 57–67.
- Zhang, J. (2017). The evolution of China’s one-child policy and its effects on family outcomes. *Journal of Economic Perspectives* 31(1), 141–160.
- Zhang, M., Y. Wang, and L. Hou (2024). Gender norms and the child penalty in china. *Journal of Economic Behavior & Organization* 221, 277–291.

Appendix

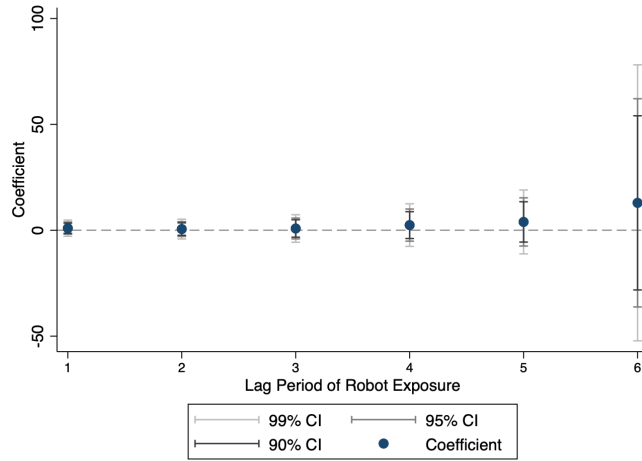
A Additional Figures and Tables

Figure A1: Estimated Effects of Robot Exposure on the Number of Children Using Different Lag Periods of Robot Exposure (CFPS)

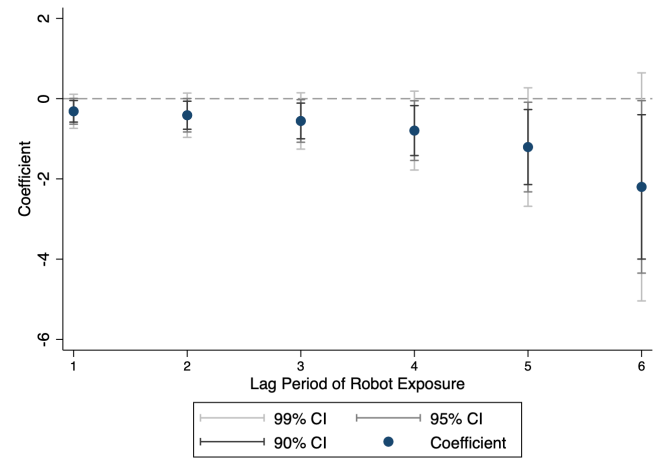


Notes: This figure presents 2SLS estimates of the effect of robot exposure on the number of children using different lag periods. The data are drawn from the CFPS (2010-2018). The regression specification is the same as in Column (4) of Table 1. The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment.

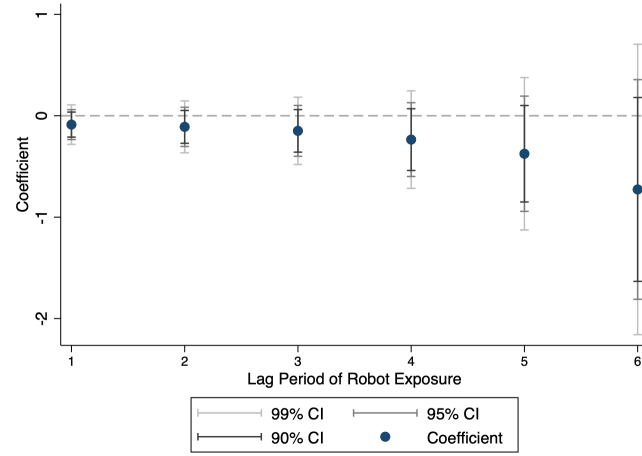
Figure A2: Estimated Effects of Robot Exposure on the Number of Children Across Different Age Stages Using Different Lag Periods of Robot Exposure (CFPS)



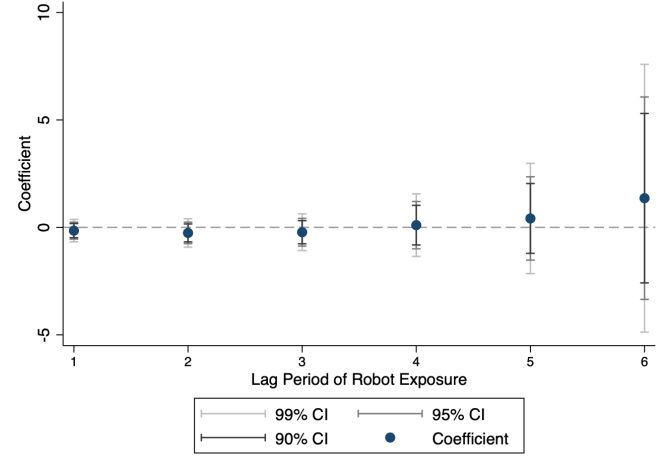
Panel A. Age 15–24



Panel B. Age 25–34



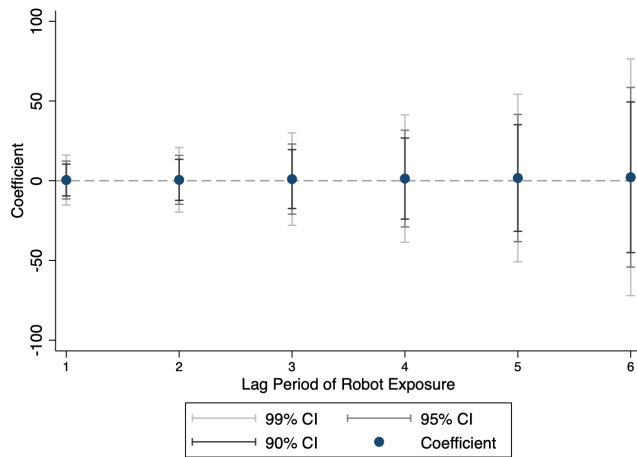
Panel C. Age 35–44



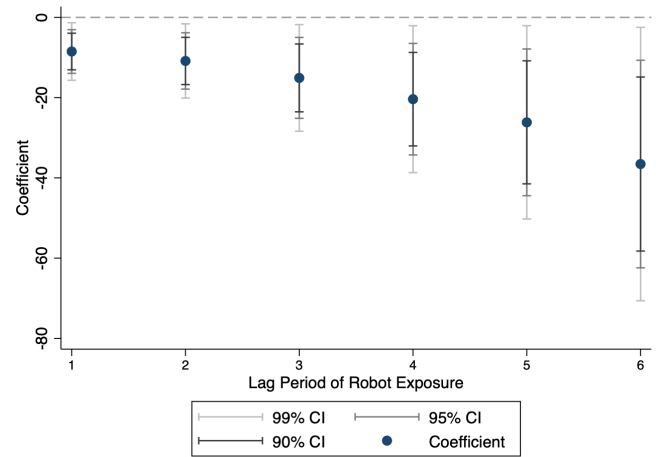
Panel D. Age 45–49

Notes: This figure presents 2SLS estimates of the effect of robot exposure on the number of children using different lag periods of robot exposure across age subgroups. The data are drawn from the CFPS (2010–2018). The regression specification follows that of Column (4) of Table 1. Panels A–D report results by age subgroup: 15–24 (Panel A), 25–34 (Panel B), 35–44 (Panel C), and 45–49 (Panel D). Each point in the figures represents the coefficient from a separate regression using a different lag of robot exposure.

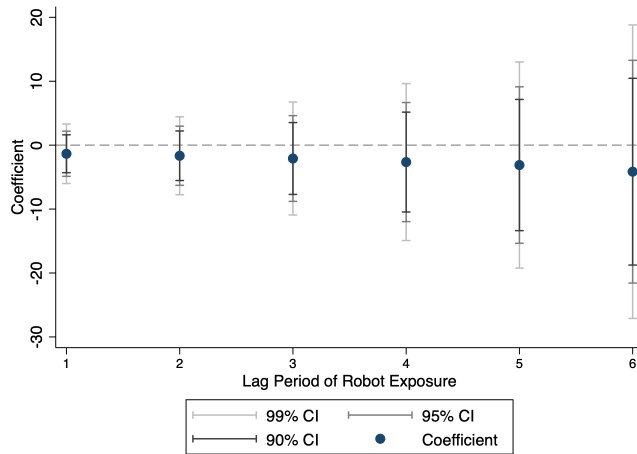
Figure A3: Estimated Effects of Robot Exposure on the Age-specific Fertility Rate Using Different Lag Periods of Robot Exposure (Census)



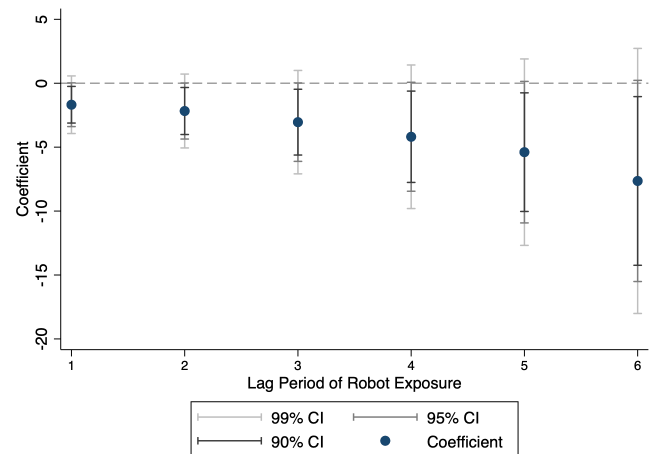
Panel A. Age 15–24



Panel B. Age 25–34



Panel C. Age 35–44



Panel D. Age 45–49

Notes: This figure presents 2SLS estimates of the effect of robot exposure on the age-specific fertility rate (ASFR) using different lag periods of robot exposure. The data are drawn from the 2010 and 2020 Censuses. The regression specification follows Column (2) in Table A2. Panels A–D show results for different age groups: 15–24 (Panel A), 25–34 (Panel B), 35–44 (Panel C), and 45–49 (Panel D). The ASFR is calculated as the number of females giving birth per 1,000 females in each age group in a given year. Each point represents the coefficient from a separate regression using a different lag of robot exposure.

Table A1: Summary Statistics

Variables	Sample	Mean	Std.	Min.	Max.
Number of Children	10,780	1.37	0.74	0	7
Child Dummy	10,780	0.939	0.24	0	1
Robot Exposure in $t - 1$	10,780	0.175	0.215	0.002	1.203
Age	10,780	36	8	17	49
Education Years	10,780	9	4	0	22
Rural Hukou	10,780	0.572	0.495	0	1
Spouse's Age	10,780	38	8	19	77
Spouse's Education Years	10,780	10	4	0	19
Employment	10,780	0.585	0.493	0	1
Monthly Working Hours	10,780	126	124	0	480
Earnings	10,780	13,740	19,688	0	390,000
Spouse's Employment	10,780	0.875	0.330	0	1
Spouse's Monthly Working Hours	10,780	187	115	0	480
Spouse's Earnings	10,780	25,957	35,301	0	1,800,000
Total Income	10,780	39,697	45,054	0	1,818,000
Wife Earns More	10,780	0.328	0.469	0	1
Share of Wife's Earnings	9,862	0.305	0.289	0	1

Notes: This table presents summary statistics for the main variables. The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment. "Child Dummy" is an indicator variable equal to 1 if the respondent has at least one child, and 0 otherwise. "Robot Exposure in $t - 1$ " captures exposure to industrial robots in the previous year, calculated using industry-level robot adoption data from the IFR. "Earnings" denotes the respondent's annual labor earnings. "Total Income" is defined as the combined annual earnings of the respondent and her spouse (i.e. Earnings + Spouse's Earnings). "Wife Earns More" is a dummy variable equal to 1 if the wife's earnings are greater than or equal to the husband's, and 0 otherwise. "Share of Wife's Earnings" represents the proportion of the wife's annual earnings in the total household income. It is missing for 918 observations with zero household income.

Table A2: Robustness Checks: Using Census Data

	Robot Exposure in $t - 1$	Total Fertility Rate
	(1)	(2)
	First stage	Second stage
Robot Exposure in $t - 1$		-0.052*
		(0.028)
IV	0.300***	
	(0.035)	
Prefecture Controls $\times$ Year Trend	Y	Y
Prefecture FE	Y	Y
Year FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	562	562
First-stage F Stat.	75.38	

Notes: This table reports the impact of robot exposure on the total fertility rate (TFR) at the prefecture level using two waves of census data in the years 2010 and 2020. The TFR aggregates age-specific fertility rates across all reproductive ages in a given year, yielding a synthetic measure of lifetime fertility under current period conditions. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Robustness Checks: Balance Test

	(1)	(2)
	Coef.	S.E.
<i>Predetermined prefecture Characteristics:</i>		
Natural Population Growth Rate	-2.497	-6.863
Total Employment	0.700	-0.463
Log of GDP	7.725	-4.782
Share of Employment in the Secondary Industry	2.528	-1.614
Share of Employment in the Tertiary Industry	2.682	-1.714
Number of Colleges	3.103	-1.945
<i>F-statistic</i>		1.34
<i>P-value</i>		0.311

Notes: The table reports coefficients and standard errors from regressing sector-level robot shocks on each sector specific weighted average of initial prefecture characteristic and year fixed effects. All regressions are weighted by the simple average of the exposure to sector-level shock across prefectures. Each coefficient comes from a separate regression. Standard errors are clustered at the sector level.

Table A4: Robustness Checks: Rotemberg Weights

Industry	Rotemberg Weight
Electronics	0.851
All Other Manufacturing Branches	0.446
Wood and Wood Products	0.336
Paper and Paper Products, Publishing & Printing	0.281
Food Products and Beverages; Tobacco Products	0.195
Metals	0.123
Textiles, Leather, Wearing Apparel	0.056
Mining and Quarrying	-0.009
Automotive	-0.057
Electricity, Gas and Water Supply	-0.070
Other Transport Equipment	-0.190
Glass, Ceramics, Stone, Mineral Products	-0.464
Plastic and Chemical Products	-0.513

Notes: This table shows the Rotemberg weight associated with each sector following the procedure introduced by Goldsmith-Pinkham et al. (2020). We calculated Rotemberg weights by sector and year. Here we report the average weight of each sector throughout the period of the analysis.

Table A5: Robustness Checks: Rotemberg Weights

	Number of Children		
	(1)	(2)	(3)
	Excluding Top 1 Sector	Excluding Top 2 Sectors	Excluding Top 3 Sectors
<i>Panel A. Second Stage</i>			
Robot Exposure in $t - 1$	-0.189*** (0.052)	-0.183*** (0.053)	-0.183*** (0.053)
<i>Panel B. First Stage</i>			
IV1	0.162*** (0.008)		
IV2		0.168*** (0.010)	
IV3			0.168*** (0.010)
First-stage F Stat.	382.18	274.29	273.63
Individual Controls	Y	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y	Y
Year FE	Y	Y	Y
Individual FE	Y	Y	Y
Province FE $\times$ Year Trend	Y	Y	Y
Observations	10,780	10,780	10,780

Notes: This table reports the effects of robot exposure on the number of children using reconstructed instruments. The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment. Column (1) reconstructs the instrument by excluding the top-ranked sector according to Rotemberg weights. Column (2) excludes the top two sectors, and Column (3) excludes the top three sectors. Individual-level controls include age, age squared, education years, rural hukou status, spouse's age, spouse's age squared, and spouse's education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Robustness Checks: Stacked First Difference

	Δ Number of Children		
	(1)	(2)	(3)
Panel A. Second Stage			
Δ Robot Exposure in $t - 1$	-0.500*** (0.153)	-0.508*** (0.154)	-0.484*** (0.140)
Panel B. First Stage			
Δ IV	0.113*** (0.010)	0.113*** (0.010)	0.128*** (0.009)
First-stage F Stat.	138.58	139.05	185.36
Prefecture Controls $\times$ Year Trend	Y	Y	Y
Year FE	Y	Y	
Province FE $\times$ Year Trend	Y	Y	
Individual Controls		Y	Y
Province FE $\times$ Year FE			Y
Observations	5,798	5,798	5,798

Notes: The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment, and who were continuously tracked across two survey waves. “ Δ Robot exposure in $t - 1$ ” is defined as the difference in robot exposure between year $t - 1$ and year $t - 3$. Δ Number of children is calculated as the number of children in year t minus that in year $t - 2$, capturing the change in fertility outcomes between two survey waves. Individual-level controls include changes in education and hukou status between the two survey waves. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A7: Robustness Checks: Controlling for Fertility Policy and Maternity Leave

	Number of Children				
	(1)	(2)	(3)	(4)	(5)
	Controlling for Fertility Policy			Controlling for Maternity Leave	
	2014–2018			2010–2018	2014–2018
Robot Exposure in $t - 1$	-0.243*** (0.080)	-0.236*** (0.081)	-0.228*** (0.081)	-0.191*** (0.052)	-0.236*** (0.082)
Non-single-child Couples Share $\times$ Post_2016	Y				
One-child Share $\times$ Post_2016		Y			
Non-single-child Couples & One-child Share $\times$ Post_2016			Y		
Maternity Leave $\times$ Post_2016				Y	Y
Individual Controls	Y	Y	Y	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Individual FE	Y	Y	Y	Y	Y
Province FE $\times$ Year Trend	Y	Y	Y	Y	Y
Observations	4,734	4,734	4,734	10,780	4,734

Notes: The data come from the CFPS. The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment. To account for the relaxation of fertility policies, we restrict the sample to observations from 2014 onward to minimize the influence of the Selective Two-Child Policy introduced in 2013 in Columns (1)–(3). Within this restricted sample, we explicitly control for the Universal Two-Child Policy introduced in 2016 using a DID approach. Column (1) controls for the interaction between the post-2016 dummy and the provincial share of families in which both spouses have siblings, using data from 2014–2018. Columns (2) and (3) alternatively include DID terms interacting the post-2016 dummy with (i) the provincial share of couples with only one child and (ii) the provincial share of non-single-child couples with only one child, respectively, using data from 2014–2018. Columns (4) and (5) control for the interaction between the post-2016 dummy and the number of days of extended maternity leave at the provincial level, using data from 2010–2018 and 2014–2018, respectively. Individual-level controls include age, age squared, education years, rural hukou status, spouse's age, spouse's age squared, and spouse's education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Robustness Checks: Province-level Maternity Leave Expansion

Province	Implementation Date	Maternity Leave (Days)		
		Before Reform	After Reform	Change
Beijing	2016.3	128	128	0
Tianjin	2016.1	128	128	0
Hebei	2016.3	143	158	15
Shanxi	2016.1	128	158	30
Inner Mongolia	2016.3	128	158	30
Liaoning	2016.3	158	158	0
Jilin	2016.3	128	158	30
Heilongjiang	2016.4	180	180	0
Shanghai	2016.2	128	128	0
Jiangsu	2016.1	128	128	0
Zhejiang	2016.1	98	128	30
Anhui	2016.1	128	158	30
Fujian	2016.2	98	158	60
Jiangxi	2016.1	128	158	30
Shandong	2016.1	158	158	0
Henan	2016.5	188	188	0
Hubei	2016.1	128	128	0
Hunan	2016.3	128	158	30
Guangdong	2016.1	113	178	65
Guangxi	2016.1	112	148	36
Hainan	2016.3	188	188	0
Chongqing	2016.4	128	128	0
Sichuan	2016.1	128	158	30
Guizhou	2016.4	128	158	30
Yunnan	2016.4	128	158	30
Tibet	2016.1	98	365	267
Shanxi	2016.8	113	168	55
Gansu	2016.4	113	180	67
Qinghai	2016.3	128	158	30
Ningxia	2016.1	138	158	20
Xinjiang	2017.7	128	158	30

Notes: This table lists the days of maternity leave and the implementation dates for all provinces in mainland China. The lengths before reform include bonus duration for late-marriage and late-childbirth. Data source: Population and Family Planning Regulations of each province.

Table A9: Robustness Checks: Controlling for Minimum Wage Policy

	Number of Children	
	(1)	(2)
	Controlling for Prefecture-year-level Minimum Wage	Controlling for Province-year-level Minimum Wage
Robot Exposure in $t - 1$	-0.181*** (0.050)	-0.189*** (0.051)
Individual Controls	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y
Year FE	Y	Y
Individual FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	10,780	10,780
First-stage F Stat.	368.01	393.39

Notes: The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment. Column (1) controls for the prefecture-year-level minimum wage. Column (2) controls for the province-year-level minimum wage. Individual-level controls include age, age squared, education years, rural hukou status, spouse's age, spouse's age squared, and spouse's education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Robustness Checks: Additional Controls

	Number of Children		
	(1)	(2)	(3)
Robot Exposure in $t - 1$	-0.189*** (0.051)	-0.199*** (0.053)	-0.304*** (0.105)
Individual Controls	Y	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y	
Year FE	Y		
Individual FE	Y	Y	Y
Province FE $\times$ Year Trend	Y		
Province FE $\times$ Year FE		Y	Y
Prefecture FE $\times$ Year Trend			Y
Observations	10,780	10,780	10,780
First-stage F Stat.	391.87	457.44	275.84

Notes: The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners’ primary occupations are in agriculture or self-employment. Column (1) reports the baseline results as shown in Column (4) of Table 1. Column (2) further controls for province-by-year fixed effects, and Column (3) additionally controls for prefecture fixed effects interacted with a year trend. Individual-level controls include age, age squared, education years, rural hukou status, spouse’s age, spouse’s age squared, and spouse’s education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A11: Robustness Checks: Impact on Marriage

	Robot Exposure in $t - 1$	Marriage (=1)
	(1)	(2)
	First stage	Second stage
Robot Exposure in $t - 1$		0.006 (0.026)
IV	0.174*** (0.013)	
Individual Controls	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y
Year FE	Y	Y
Individual FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	35,160	35,160
First-stage F Stat.	172.80	

Notes: This table presents estimates of the effect of robot exposure on marriage (married = 1). The data are drawn from the CFPS for the period 2010–2018. The sample consists of individuals aged 16–49 whose primary occupations are not in agriculture or self-employment, and who are observed in at least two survey waves. Individual-level controls include age, age squared, education years, rural hukou status, spouse’s age, spouse’s age squared, and spouse’s education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: Robustness Checks: Extending the Sample to Agricultural and Self-employed Workers

	Robot Exposure in $t - 1$	Number of Children
	(1)	(2)
	First stage	Second stage
Robot Exposure in $t - 1$		-0.094*
		(0.056)
IV	0.162***	
	(0.007)	
Individual Controls	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y
Year FE	Y	Y
Individual FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	21,641	21,641
First-stage F Stat.	509.57	

Notes: This table presents estimates of the effect of robot exposure on the number of children. The data are drawn from the CFPS for the period 2010–2018. The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data. Individual-level controls include age, age squared, education years, rural hukou status, spouse’s age, spouse’s age squared, and spouse’s education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A13: Robustness Checks: Using an Alternative Instrument and Altering the Sector Coverage

	Number of Children	
	(1)	(2)
	Altering IV's Country Coverage	Altering the Sector Coverage
Panel A: Second Stage		
Robot Exposure in $t - 1$	-0.174*** (0.048)	
Manufacturing Robot Exposure in $t - 1$		-0.189*** (0.051)
Panel B: First Stage		
IV_EU_USA	0.442*** (0.018)	
Manufacturing IV		0.161*** (0.008)
First-stage F Stat.	611.19	392.35
Individual Controls	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y
Year FE	Y	Y
Individual FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	10,780	10,780

Notes: The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment. Column (1) reports results using an alternative instrument constructed from robot adoption in Europe and the United States. In Column (2), both robot exposure and the instrument are calculated using manufacturing sectors only. Individual-level controls include age, age squared, education years, rural hukou status, spouse's age, spouse's age squared, and spouse's education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A14: Robustness Checks: Using CFPS 2010-2020

	Robot Exposure in $t - 1$	Number of Children
	(1)	(2)
	First Stage	Second Stage
Robot Exposure in $t - 1$		-0.139** (0.063)
IV	0.170*** (0.011)	
Individual Controls	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y
Year FE	Y	Y
Individual FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	11,908	11,908
First-stage F Stat.	222.79	

Notes: The data come from the CFPS (2010–2020). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment. Individual-level controls include age, age squared, education years, rural hukou status, spouse's age, spouse's age squared, and spouse's education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A15: Robustness Checks: Clustering at Different Levels

	Number of Children	
	(1)	(2)
	Clustered at the Province Level	Clustered at the Individual Level
Robot Exposure in $t - 1$	-0.189*** (0.056)	-0.189*** (0.057)
Individual Controls	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y
Year FE	Y	Y
Individual FE	Y	Y
Province FE $\times$ Year Trend	Y	Y
Observations	10,780	10,780
First-stage F Stat.	175.68	8821.62

Notes: The data come from the CFPS (2010–2018). The sample includes non-single females aged 17–49 (married or cohabiting) whose partners are observed in the data, and for whom neither their own nor their partners' primary occupations are in agriculture or self-employment. Individual-level controls include age, age squared, education years, rural hukou status, spouse's age, spouse's age squared, and spouse's education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. In Column (1), standard errors are clustered at the province level, while in Column (2) they are clustered at the individual level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A16: Alternative Channels: Quantity–quality Trade-off

	Log (Education Expense)	Chinese Scores	Math Scores
	(1)	(2)	(3)
Robot Exposure in $t - 1$	-0.054 (0.960)	0.481 (0.316)	0.201 (0.310)
Individual Controls	Y	Y	Y
Prefecture Controls $\times$ Year Trend	Y	Y	Y
Year FE	Y	Y	Y
Individual FE	Y	Y	Y
Province FE $\times$ Year Trend	Y	Y	Y
Observations	7,637	3,680	3,680
First-stage F Stat.	113.04	65.30	65.30

Notes: The data come from the CFPS (2010–2018). Column (1) presents the baseline sample’s children aged 0–15 who were tracked for at least two waves between 2010 and 2018, with the dependent variable being the logarithm of total educational expenditure over the past 12 months. The estimated effect is statistically insignificant, suggesting no impact on parental educational investment. Columns (2) –(3) focus on the baseline sample’s children attending school (aged 6–15), with the dependent variables being parental assessments of the child’s performance in Chinese and math, respectively. The scores range from 1 to 4, corresponding to “poor,” “average,” “good,” and “excellent.” The estimates indicate no statistically significant effects on either subject. Individual controls include child’s rural hukou, child’s gender, mother’s age, mother’s employment status and mother’s education years. Prefecture-level controls include the natural population growth rate, the share of employment in the tertiary sector, the number of employees at year-end, and the logarithm of GDP in 2010. Standard errors are clustered at the prefecture level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B Model Derivations

Derivation of the unit cost.

Let $\alpha \equiv (\sigma - 1)/\sigma \in (-\infty, 0)$. Then (3) can be written as

$$n = (v_m^\alpha t_m^\alpha + v_f^\alpha t_f^\alpha)^{1/\alpha}.$$

To obtain the unit cost, set $n = 1$ and solve

$$\min_{t_m, t_f} w_m t_m + w_f t_f \quad \text{s.t.} \quad (v_m^\alpha t_m^\alpha + v_f^\alpha t_f^\alpha)^{1/\alpha} \geq 1.$$

The Lagrangian is

$$\mathcal{L} = w_m t_m + w_f t_f + \mu \left(1 - (v_m^\alpha t_m^\alpha + v_f^\alpha t_f^\alpha)^{1/\alpha} \right).$$

The FOCs imply (for an interior solution)

$$\frac{w_m}{w_f} = \frac{v_m^\alpha t_m^{\alpha-1}}{v_f^\alpha t_f^{\alpha-1}} \quad \Rightarrow \quad \left(\frac{t_m}{t_f} \right)^{\alpha-1} = \frac{w_m}{w_f} \left(\frac{v_f}{v_m} \right)^\alpha.$$

Using $\alpha - 1 = -1/\sigma$, this gives a standard CES demand system. Let $q_j \equiv w_j/v_j$ for $j \in \{m, f\}$.

The conditional demands for one unit of child output satisfy $v_j t_j = (p/q_j)^\sigma$. Substituting these demands into the unit-output constraint gives

$$p^{1-\sigma} = q_m^{1-\sigma} + q_f^{1-\sigma},$$

and therefore

$$p(w_m, w_f) = \left[\left(\frac{w_m}{v_m} \right)^{1-\sigma} + \left(\frac{w_f}{v_f} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}.$$

By homogeneity of degree one in n (CRS), $C(n, w_m, w_f) = n p(w_m, w_f)$.

Cost shares (Shephard's lemma).

The unit-cost shares are

$$\theta_m \equiv \frac{\partial \ln p}{\partial \ln w_m} = \frac{\left(\frac{w_m}{v_m}\right)^{1-\sigma}}{\left(\frac{w_m}{v_m}\right)^{1-\sigma} + \left(\frac{w_f}{v_f}\right)^{1-\sigma}}, \quad \theta_f = 1 - \theta_m.$$

Under $\sigma < 1$, both shares are interior: $\theta_m, \theta_f \in (0, 1)$.

Microfoundation of β_f^{eff} .

Take fertility n as given and consider Stage 2. In the non-cooperative default, each spouse consumes their own earnings net of the childcare burden assigned by the default cost share τ_f :

$$\begin{aligned} \bar{u}_f(n, \tau_f) &= \ln[w_f - \tau_f p(w_m, w_f) n] + \beta_f \ln n, \\ \bar{u}_m(n, \tau_f) &= \ln[w_m - (1 - \tau_f) p(w_m, w_f) n] + \beta_m \ln n. \end{aligned}$$

Define the husband's outside-option advantage:

$$\eta(n, \tau_f) \equiv \bar{u}_m(n, \tau_f) - \bar{u}_f(n, \tau_f) = \ln \left[\frac{w_m - (1 - \tau_f)pn}{w_f - \tau_f pn} \right] + (\beta_m - \beta_f) \ln n.$$

Differentiating with respect to τ_f :

$$\frac{\partial \eta}{\partial \tau_f} = \frac{pn}{w_m - (1 - \tau_f)pn} + \frac{pn}{w_f - \tau_f pn} > 0.$$

A higher default childcare burden worsens the wife's fallback and improves the husband's. Let $\tilde{\lambda}(n, \tau_f) = \tilde{\Lambda}(\eta(n, \tau_f))$ with $\tilde{\Lambda}' > 0$ denote the male renegotiation weight in Stage 2. The wife's Stage-2 indirect utility is then

$$V_f(n, \tau_f) = \ln[(1 - \tilde{\lambda}(n, \tau_f))(S - pn)] + \beta_f \ln n = \ln[1 - \tilde{\lambda}(n, \tau_f)] + \ln(S - pn) + \beta_f \ln n.$$

Since $\partial \tilde{\lambda} / \partial \tau_f > 0$, a higher τ_f lowers the wife's consumption share $1 - \tilde{\lambda}$ and hence her continuation value after a birth.

Moreover, a larger family size n worsens the wife's fallback position (the net income $w_f - \tau_f pn$ in $\bar{u}_f(n, \tau_f)$ falls), which further raises $\tilde{\lambda}$. This interaction between τ_f and n creates a commitment friction: the term $\ln[1 - \tilde{\lambda}(n, \tau_f)]$ effectively reduces the wife's willingness to have children beyond what β_f alone would imply.

A second-order Taylor expansion of $\ln[1 - \tilde{\lambda}(n, \tau_f)]$ in the variables $(\ln n, \tau_f)$ around $(\bar{n}, \bar{\tau}_f)$, retaining only the cross-partial term and normalising the baseline so that the pure partials $C_1 \equiv \partial \ln[1 - \tilde{\lambda}] / \partial \ln n|_{\bar{n}, \bar{\tau}_f}$ and $C_2 \equiv \partial \ln[1 - \tilde{\lambda}] / \partial \tau_f|_{\bar{n}, \bar{\tau}_f}$ are absorbed into the constants K and β_f , yields⁹

$$\ln[1 - \tilde{\lambda}(n, \tau_f)] \approx K - \gamma \tau_f \ln n,$$

where K is a constant and

$$\gamma \equiv - \left. \frac{\partial^2 \ln[1 - \tilde{\lambda}(n, \tau_f)]}{\partial \tau_f \partial \ln n} \right|_{(\bar{n}, \bar{\tau}_f)} > 0.$$

The positivity of γ is structural: a larger default childcare burden makes the wife's outside option deteriorate more steeply with fertility, which lowers her renegotiation share after children are born. Substituting the approximation into $V_f(n, \tau_f)$ gives

$$V_f(n, \tau_f) \approx K + \ln(S - pn) + [\beta_f - \gamma \tau_f] \ln n.$$

Hence the wife's Stage-1 problem is isomorphic to one with effective fertility preference $\beta_f^{\text{eff}} = \beta_f - \gamma \tau_f$, establishing equation (4).

⁹The normalisation $C_1 = C_2 = 0$ amounts to redefining the baseline $(\bar{n}, \bar{\tau}_f)$ as the point where the wife's Stage-2 first-order conditions are satisfied, so that deviations from baseline enter only through the cross-partial. This is without loss of generality: γ is identified from the cross-partial $\partial^2 / \partial \tau_f \partial \ln n$, which is invariant to the choice of baseline levels.

Derivation of closed-form fertility (6).

Fix n . Maximizing (5) over (c_m, c_f) subject to $c_m + c_f = S - pn$ yields $c_m = \lambda(S - pn)$ and $c_f = (1 - \lambda)(S - pn)$. Substituting reduces the problem to

$$\max_n \ln(S - pn) + \bar{\beta} \ln n.$$

The FOC is $-p/(S - pn) + \bar{\beta}/n = 0$, which implies $\bar{\beta}S = (1 + \bar{\beta})pn$ and hence (6).

Channel-by-channel algebra.

From (6), $\ln n^* = \ln(\bar{\beta}/(1 + \bar{\beta})) + \ln S - \ln p$. The bargaining term satisfies the following requirements:

$$\frac{d}{d\rho} \ln \left(\frac{\bar{\beta}}{1 + \bar{\beta}} \right) = \frac{1}{1 + \bar{\beta}} \cdot \frac{d \ln \bar{\beta}}{d\rho}.$$

The income channel: let $s_m \equiv w_m/S$ and $s_f \equiv w_f/S$. Under (7), $d \ln S/d\rho = -(\delta_m s_m + \delta_f s_f) \leq 0$.

The price channel: using Shephard's lemma, $d \ln p/d\rho = -(\theta_m \delta_m + \theta_f \delta_f)$, so $-d \ln p/d\rho > 0$.

The bargaining channel: from $\bar{\beta} = \lambda \beta_m + (1 - \lambda) \beta_f^{\text{eff}}$ and $d\tau_f/d\rho = 0$, we get

$$\frac{d \ln \bar{\beta}}{d\rho} = \frac{\beta_m - \beta_f^{\text{eff}}}{\bar{\beta}} \cdot \frac{d\lambda}{d\rho}.$$

Combining with Lemma 1:

$$\frac{d}{d\rho} \ln \left(\frac{\bar{\beta}}{1 + \bar{\beta}} \right) = -\frac{\beta_m - \beta_f^{\text{eff}}}{\bar{\beta}(1 + \bar{\beta})} \kappa(r) (\delta_m - \delta_f) < 0.$$

Proof of Proposition 1.

Define

$$N(\tau_f) \equiv \beta_m - \beta_f^{\text{eff}} = \beta_m - \beta_f + \gamma \tau_f, \quad D(\tau_f) \equiv \bar{\beta}(1 + \bar{\beta}).$$

The bargaining term in (9) is

$$B(\tau_f) \equiv \frac{N(\tau_f)}{D(\tau_f)} \kappa(r) (\delta_m - \delta_f).$$

We have $\partial N / \partial \tau_f = \gamma > 0$. From (5),

$$\frac{\partial \bar{\beta}}{\partial \tau_f} = -(1 - \lambda)\gamma < 0,$$

so $\partial D / \partial \tau_f = (1 + 2\bar{\beta}) \cdot \partial \bar{\beta} / \partial \tau_f < 0$. Hence $N(\tau_f)$ is increasing and $D(\tau_f)$ is decreasing in τ_f , implying that $B(\tau_f)$ is strictly increasing in τ_f . Since the full derivative in (9) equals the income and price components minus $B(\tau_f)$, and those two components do not vary with τ_f in this specification,

$$\frac{\partial}{\partial \tau_f} \left(\frac{d \ln n^*}{d \rho} \right) = -\frac{\partial B}{\partial \tau_f} < 0.$$

If the total effect is negative, this also implies

$$\frac{\partial}{\partial \tau_f} \left| \frac{d \ln n^*}{d \rho} \right| = \frac{\partial B}{\partial \tau_f} > 0.$$

Proof of Proposition 2.

A wider spousal preference gap operates through women's systematically lower desired fertility, i.e., a lower β_f holding β_m fixed, so that $\beta_m - \beta_f$ rises. Using the notation of the previous proof with $N \equiv \beta_m - \beta_f^{\text{eff}} = \beta_m - \beta_f + \gamma \tau_f$ and $D \equiv \bar{\beta}(1 + \bar{\beta})$,

$$\frac{\partial N}{\partial \beta_f} = -1, \quad \frac{\partial \bar{\beta}}{\partial \beta_f} = (1 - \lambda) > 0, \quad \frac{\partial D}{\partial \beta_f} = (1 + 2\bar{\beta})(1 - \lambda) > 0.$$

Hence

$$\frac{\partial B}{\partial \beta_f} = \frac{(\partial N / \partial \beta_f)D - N(\partial D / \partial \beta_f)}{D^2} = \frac{-D - N(1 + 2\bar{\beta})(1 - \lambda)}{D^2} < 0,$$

so B is strictly decreasing in β_f and strictly increasing in the gap $\beta_m - \beta_f$. Since the income and opportunity-cost components of (9) do not depend on preferences,

$$\frac{\partial}{\partial (\beta_m - \beta_f)} \left(\frac{d \ln n^*}{d \rho} \right) = -\frac{\partial B}{\partial (\beta_m - \beta_f)} < 0.$$